Estimating Intergenerational Returns to Medical Care: New Evidence from At-Risk Newborns*

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Abstract

We examine whether intensive early-life government-funded interventions targeted to at-risk newborns are transmitted intergenerationally. Using a regression discontinuity design and administrative data from Chile we follow women up to the age of 25, and document the surprising fact that children of individuals who were treated at birth have worse indicators of health at birth a generation later. We suggest this owes to selective fertility, finding that marginally treated individuals are substantially more likely to give birth. These new stylised facts suggest that in certain circumstances, the long-term implications of public investments within family lineages may be quite different to their short-term implications, placing more weight on necessary reinforcing interventions.

JEL Codes: I11; I18; J13; H51; O15.
Keywords: Early life interventions; intergenerational mobility; parental investments; fertility; health care provision.

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1 Introduction

Returns to early life public investment programmes accrue over life. How generalised is this accrual, for how long does it last, and over what dimensions do investments accrue? These questions have important implications for policy design related to optimal welfare investments over the life-cycle (Cunha et al., 2010; Currie and Rossin-Slater, 2015). Conceivably, if early-life circumstances shape later life outcomes, policies which affect such investments could have considerable intergenerational implications (Mogstad and Torsvik, 2022), suggesting impacts may echo well beyond the period of policy receipt.

In this paper we seek to understand these questions, studying whether targeted and intensive publicly-funded medical interventions in very low birth weight (VLBW) children immediately after birth are transmitted onto the outcomes of *future* generations. Using a multigenerational linked database of the universe of births in Chile and a regression discontinuity (RD) design, we examine whether women who were born weighing just below a specific (1,500 gram) treatment cut-off go on to have babies who have greater measurable stocks of health at birth a generation later. We trace out the impacts of an individual's early life treatment on their future interactions with the health care system, the composition of their own family during childhood, their fertility behaviour upon maturity, and ultimately, the intergenerational transmission of intensive medical care receipt to measures of their children's health and well-being.

Large public health programmes targeted to children with poor birth outcomes result in immediate improvements in health and survival (Almond et al., 2010; Bharadwaj et al., 2013; Chyn et al., 2021), which impact educational outcomes in childhood and adolescence (Bharadwaj et al., 2013; Daysal et al., 2022), reduce future reliance on public social safety net programmes (Chyn et al., 2021) and positively affect other members of the family including siblings and mothers over both the short and long term (Daysal et al., 2022).[1] More generally, there is substantial evidence from a broad literature in economics documenting intergenerational transmission of health at birth (Currie and Moretti, 2007; Nilsson, 2017; Giuntella et al., 2019; Lahti-Pulkkinen et al., 2017; Royer, 2009, for example). At the same time, the way individuals interact with health policies, and the degree to which treatment receipt can shape future outcomes across the life course, is complex and multi-faceted. Both Currie and Rossin-Slater (2015) and Wüst (2022), reviewing a broad range of studies, demonstrate that early life policies have appreciable impacts across multiple later

[1] These studies, based on a similar RD design as that used in this paper, point to VLBW cut-offs as significant determinants of well-being for individuals during early life and childhood. The study of Bharadwaj et al. (2013) considers the first generation using the same data and the same context studied in this paper. Their results document a strong and enduring 'first-stage' to the longer term and intergenerational context we study in this paper. As we discuss in the methods and results section of this paper, we document that their earlier results still hold when working with a much extended sample and recent advances in optimal RD designs (as has been documented in other recent RD papers of this nature such as Daysal et al. (2022)).

life domains including adult health, educational attainment, labour market outcomes, and other measures of socio-economic status. If neonatal and early life health programmes lift up the life courses of their original recipients, and in turn spillover to future generations, this may suggest that the already large benefits of such programmes could be a (considerable) lower bound. However, if families or individuals change their behaviour as a downstream result of medical intervention (or lack of intervention), or if medical intervention shapes intermediate outcomes potentially shifting the life courses of individuals, the intergenerational returns to medical care may be entirely different to returns to the first generation. Indeed, intergenerational returns could even be negative if families make compensating investments in less treated individuals thus overwhelming initial investments received by more intensively treated individuals, or if medical receipt shapes future individual behaviours in other unexpected ways.[2]

In this study we trace out impacts of the receipt of intensive medical care very early in childhood on outcomes in childhood, adolescence, early adulthood, and spillovers into future generations. Linking comprehensive microdata registries, we are able to observe all births in Chile between 1992-2018, their future inpatient hospitalisations, and their future fertility histories, and, for those women that go on to have *their own* births, we observe the early life health stocks and survival records for their children. Our results show that despite relatively large effects of the programme on infant mortality and days of hospitalisation in the first generation suggesting a substantive first generation impact (Bharadwaj et al., 2013), there is clearly no positive intergenerational transmission to the health outcomes at birth of the *children* of individuals who were treated at birth. Indeed, more surprisingly, we observe quite clear negative transmissions of intensive medical treatment receipt at birth to the health outcomes of second generation of children, especially when studying outcomes very low in the distribution of health at birth, such as the likelihood that a child is born prematurely, or with a very low birth weight.

We consider a number of channels which may explain the reversal of impacts across generations. These include selection in terms of fertility, changes in fertility timing or the composition of individuals giving birth, survival selection upon treatment receipt in the first generation, and (first generation) parental reinforcing behaviours overwhelming initial positive effects of intensive early life health treatments. We additionally consider the likelihood that the lack of observed effects owes to low statistical power. In conducting these tests, we find clear evidence of strong policy effects on fertility many years after treatment is received. These results suggest a chain of policy impacts, whereby individuals who receive intensive medical treatment are

[2]There is evidence, for example, that early life health shocks may interact with future events and investments, potentially magnifying the impact of initial shocks (Duque et al., 2019). In general, family circumstance has large and pervasive intergenerational implications (see for example Eshaghnia et al. (2022)), which in the absence of some exogenous shock, may make pinning down the causal intergenerational impact of early life health care receipt difficult.

more able to take births to term, with subsequent negative implications on average health stocks of affected cohorts of children.

While these findings are based on rich microdata, it is important to note that there are limits to these data. Most clearly, we are only able to match mothers (and not fathers) with their children, and, given that high-quality birth records only exist in our (developing country) sample from 1992 onwards, we are limited with the maximum age-range of potential second generation mothers which we observe. More specifically, we are limited to focus on women up to a maximum age of 25 years (see discussion in Appendix A). Thus, while we work with the full sample of over 6.5 million births occurring in our study period, we view our findings as informative for young mothers, and do not make claims to broader external validity outside of this group. Nevertheless, the results and stylized facts we document suggest revisions are required to the standard understanding of intergenerational transfers when policy can shape outcomes on both extensive (fertility) and intensive (health at birth) outcomes. These results provide a counter-example, in this particular setting and sample, to the broad and growing literature which documents positive intergenerational transfers, at least if only intensive margin outcomes are observed.

This result is novel, given that the overwhelming body of the literature suggests that there are positive intergenerational spillovers in health and in socioeconomic measures more broadly, and that such positive spillovers are observable in mean outcomes of the population. For example, in examining health shocks, Lee (2014) points to intergenerational impacts still persistent in grandchildren due to their grandmother's exposure to a violent uprising while pregnant, with similar patterns observed when considering in utero declines in pollution exposure as a result of the US Clean Air Act (Colmer and Voorheis, 2020). In a recent study East et al. (2023) document important positive intergenerational spillovers owing to Medicaid coverage in the prenatal period. More broadly, there is evidence to suggest that maternal exposure to stressful events in adolescence and adulthood is passed across generations (Akresh et al., 2021). Similarly, Almond et al. (2012) document intergenerational transfer of exposure to disease across generations of mothers. Bhalotra and Rawlings (2011, 2013) document significant gradients in the exposure to health shocks of children based on the health of their mothers, additionally pointing to important channels of intergenerational transmission of health. These first-generation exposures have also been shown to be reflected in later life health outcomes of second generation children (Venkataramani, 2011; Bencsik et al., 2023), to be observed in long-term measures such as life expectancy (Black et al., 2022), to be pervasive across country income levels and settings (Halliday et al., 2021; Chang et al., 2024; Halliday, 2020), and to endure across more than two generations (Costa, 2021). A rich stream of emerging literature has shown intergenerational transfers in

education and well-being, with important implications on the accrual of inequality over generations.[3]

This paper contributes to a range of literatures. It joins a number of recent studies in documenting intergenerational transmission of early life health indicators. It also contributes to a literature considering the intergenerational transfer of exposures to particular events or environments across cohorts. It additionally joins a large literature considering the returns to public health programmes. This is one of only a small number of studies to consider intergenerational implications of individual-level exposure to a large public health policy, allowing for the estimation of long-term returns of such programmes.[4] The study closest to ours in this regard is East et al. (2023): an impressive multigenerational study of Medicaid, finding positive intergenerational spillovers in the US. Importantly, our study and their study work with a broadly similar age range of second generation mothers (though of course a quite different exposure variable), suggesting a useful contrast to our proof of concept that *in some settings* intergenerational spillovers can be negative when studying observed health outcomes. Perhaps the main contribution of this study, and where it expands knowledge in each of the literatures discussed above, is in documenting that the marginal returns to medical care in one generation can have considerable long-term implications, improving the outcomes of individuals in the first generation while at the same time bequeathing weaker health stocks to the following generation when a selective fertility channel is operating.[5]

The structure of the paper is as follows. In Section 2 we discuss the context studied here and the nature of the VLBW assignment threshold as a determinant of early life medical care. Section 3 discusses the intergenerational linked administrative data generated for this study, and Section 4 discusses how this is used to estimate impacts of exposure based on an RD design, additionally presenting a simple model to understand the full content of RD estimates in this context. Section 5 provides results as well as identification checks, discussing both mechanisms which could explain the observed results, and their implications on how we should conceive the (long-term) returns to medical care. Section 6 briefly concludes.

[3] Two studies have also considered the interaction of educational and health policies: Rossin-Slater and Wüst (2020) examining nurse visits and pre-school, and Barr and Gibbs (2022) who examine Head Start which includes health screenings, both finding substantial positive intergenerational effects.

[4] There are a number of studies which consider how exposure to policies can impact educational or labour market of exposed individuals lifting up trajectories of disadvantaged individuals, in this way breaking intergenerational cycles of disadvantage. Examples of such policies, particularly in the Scandinavian context, are discussed by Wüst (2022); see for example Bütikofer et al. (2019). What is different in this paper is the consideration of how exposure to a health policy is transmitted into *future* generations.

[5] This paper also contributes, in a limited way, to a literature on the replicability crisis in social sciences, and concerns with the use of non-public data. For a particular study (Bharadwaj et al., 2013) when previously private data was subsequently published without restrictions, we were able to substantively replicate the findings of the original paper's private data, starting from scratch in data collation and generation. Further, updating results based on technical advances in the intervening period since the paper's publication, and additionally updating to include a substantially longer time-frame which more than duplicated the original sample points to results that are entirely consistent with those in the originally published research.

2 Background and Medical Care Regimes

Birth registration in Chile is universal, and the large majority of births (over 99%) are attended in public hospitals or private clinics which follow national-level protocols set by the Ministry of Health. Births are overwhelmingly attended by doctors and/or midwives. As laid out in Bharadwaj et al. (2013), national-level guidelines were set in 1991 by a national committee to standardise treatments at Neonatal Intensive Care Units (NICUs) in the country, which exist in each of Chile's 16 regions. A particular concern at this point was the high rate of infant mortality among very low birth weight infants (weights below 1,500 grams). Treatment protocols often explicitly mention 1,500 grams as a treatment cut-off, and these births undergo a systematic treatment protocol with follow-up procedures (Hübner et al., 2009).

Bharadwaj et al. (2013), pointing to Gonzalez et al. (2006); Jiménez and Romero (2007); Palomino et al. (2005) note a number of explicit treatment cut-offs which are documented at 1,500 grams. This includes the use of artificial lung surfactant, a complementary nutrition programme providing specialised supplementation (*PNAC prematuro*), and a health care reform (*AUGE*) defining neonatal follow-up appointments to screen for particular pathologies which are targeted at infants born at less than 1,500 grams. These criteria are explicitly recognised also in later policy documents, such as the National neonatal guidebook issued to medical practitioners (Mena Nanning et al., 2005),[6] and when formally included as requirements for accessing policies (such as that indicated in the *AUGE* reform), these require children to have weights *strictly below* 1,500 grams, implying that those who weigh 1,500 grams will not classify for treatment.

Discontinuous treatment assignments at 1,500 grams have been apparent in official clinical guidelines for an extended period of time. The 2005 Neonatal Guide, referring to actions taken in the previous decade, notes policies targeted to reduce rates of morbidity among VLBW infants such as the national surfactant programme (and other programmes to standardise use of drugs such as Indometacin), programmes to follow infants over an extended period of time post-birth, the regionalisation of neonatal services (Mena Nanning et al., 2005), and recommendations that all newborns weighing less than 1,500 grams should receive diagnostic ultrasounds to examine intracranial haemorrhage, diagnostic tests for retinopathy, yearly specialised ophthalmological follow-ups, and a range of other specialised treatment courses in cases of particular diagnoses. There is evidence that this discontinuity remains highly relevant even late in the period under study. For example, the Clinical Manual for Neonatal Care written in 2016 for one of Santiago's large public hospitals with a high complexity neonatal care unit lists a large number of specific treatments which

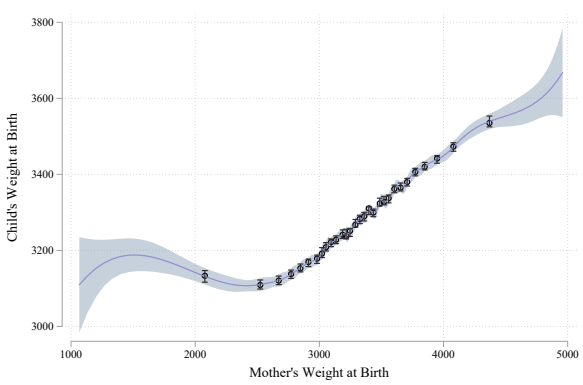
[6] Searches in this document suggest 24 occurrences of the use of a 1,500 grams, while other arbitrary cut-points are not similarly prevalent (e.g., 4 mentions of 2,000 grams and only 1 mention of 2,500 grams which is the cut-off for definitions of low birth weight (LBW)).

neonates should receive if they weigh under 1,500 grams. These include specialised procedures for temperature maintenance at birth, a delayed clamping of the umbilical cord to allow for greater blood flow to the baby following birth, the required presence of two specialists trained in resuscitation, specific formulae for ventilation and nasal air flow masks, and direct transfer to neonatal intensive care units. An entire chapter is dedicated to these procedures in the hospital's clinical guides (Mulhaüsen Muñoz and González, 2016). No similar such guidelines exist for other distributional points of birth weight.

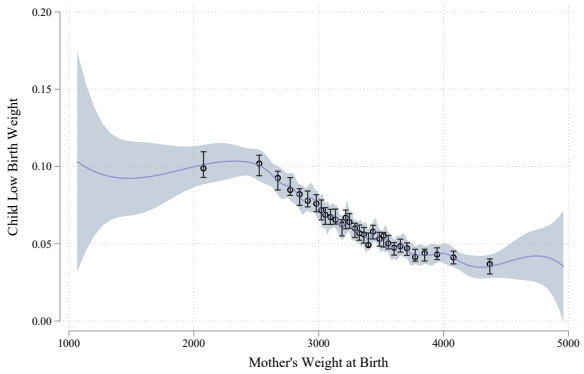
Many of these policies are additionally targeted at children who are born at less than 32 or 33 weeks of gestation, suggesting that the discontinuity may only bind for babies born at 32 weeks or above. Indeed, this is an argument made by Bharadwaj et al. (2013), who note that often policy documents or formal selection criteria for programmes also note extreme prematurity along with very low birth weight. While this is often the case, it appears that it is not uniformly so. For example, in recent technical guidelines, a number of treatments are indicated as owing exclusively to the 1,500 gram threshold, and not gestational limits (e.g., temperature regulation procedures discussed in Mulhaüsen Muñoz and González (2016)). As we discuss later in the paper, we also observe evidence suggesting elevated rates of hospitalisation for babies born just below 1,500 grams even among individuals born at 32 weeks or earlier. Thus, while it is clear that treatment rules will bind most cleanly for babies born below 1,500 grams but at 32 weeks of gestation or greater, it is not necessarily clear that no discontinuity will exist around the 1,500 gram cut-off for babies born at below 32 weeks. Thus while we generally follow Bharadwaj et al. (2013) in focusing on births at above 32 weeks of gestation, we additionally show that results are robust to relaxing this sample restriction.

In considering intergenerational links in health at birth, a stylised fact is that there is considerable, but not complete, closure of gaps in health at birth across generations. In Figure 1 we document, using microdata on all intergenerational links between births in Chile occurring after 1992, that while there is a clear tendency of lower birth weight children to go on to have births which themselves have poorer average health outcomes at birth, the intergenerational links are considerably tempered, with all binned averages located above 3,000 grams. These binned scatter plots suggest that on average, individuals weighing as little as 1,500 grams have children with birth weights of over 3,000 grams, with this value rising steadily, reaching about 3,400 grams on average for individuals who weighed 4,000 grams at birth. Thus, a considerable reversion to the mean is observed, suggesting the existence of some intergenerational dependence in health measures at birth, but with considerably cushioned impacts. Similar patterns are observed when considering low birth weight indicators, with children of LBW mothers being more likely to themselves be LBW, but nevertheless, 90% of individuals whose mothers were LBW "graduate out" of this status in the next generation. This result lines

Figure 1: Intergenerational Transmission of Early Life Health Measures



(a) Maternal birth weight and children's birth weight



(b) Maternal birth weight and child's low birth weight status

Notes: Each sub-plot documents average outcomes of individuals born to mothers whose birth weight is indicated on the horizontal axis. Panel (a) considers the birth weight of second-generation children, while panel (b) considers the average proportion of second-generation children who are low birth weight (weight $<$ 2500 grams). Optimally spaced bins and their confidence intervals are documented as black points and error bands. A cubic B-spline and its 99% confidence interval is overlaid as a continual solid line and shaded area. Optimal definitions and recommendations follow Cattaneo et al. (2019a).

up with a broader literature focused on intergenerational transmission of health in adulthood (*i.e.* between parents and their adult children). Intergenerational health associations of 0.2-0.3 are typically found when considering both physical and mental health across a range of settings (Halliday, 2020; Halliday et al., 2021; Chang et al., 2024). The results observed for birth weight here consistent with the lower end of this range, suggesting intergenerational associations of around 0.2.[7]

3 Data

We generate matched microdata covering all the 6,617,637 births occurring in Chile between 1992 and 2018. These births are matched to their future survival history, inpatient hospitalisation records, and any of their own births occurring in the future. In the case of any future births, we observe the birth outcomes, and survival history of their children. For each birth we observe both child- and parent-level measures. Thus, these data are longitudinal covering up to 3 generations: characteristics of the mother and father of children born in generation 1, as well as characteristics of any births which occur later to children of women from generation 1. These generation 2 births allow us to observe the characteristics of babies from generation 1 when they go on to have their own children. In total, of the 6,617,637 births occurring between 1992-2018, 6,147,623 have valid information for all relevant variables and meet maternal age criteria discussed below.

[7] This can be appreciated in Figure 1(a): when shifting from mother's weight of 2500 to 4000g, average children's weight shifts from around 3150 to 3500g, an association of 0.21.

Of those births, 3,010,251 (48.97%) are girls, 420,389 of whom go on to have their own future live birth. For these births we thus observe their outcomes at birth and in early life, their mother's outcomes at birth, early life and at the date of their child's birth, and their grandparents' characteristics at the moment of their child birth.

We follow Lu and Vogl (2022) in referring to an intergenerational link as a lineage. We observe lineages which consist of mothers and their children, where mothers are observed both when they were born, in which case information on their mother (the "grandmother") is observed, and if they give birth, are additionally observed at this time, in which case information on their children is observed. We lay out this nomenclature, and the structure of lineages in our data in Appendix Figure A1. Birth certificate records which are provided by the Ministry of Health contain each mother's national identification number, but do not contain the father's national identification number. For this reason, we are only able to work with links between a *mother's* health across generations as matching boys to their children is impossible. Available evidence suggests that father's health is also transmitted to future generations (Giuntella et al., 2019), though to a lesser degree than mothers (Chang et al., 2024), which is a point we return to when considering total policy effects in Section 5.1.

Birth, Death and Hospitalisation Data Birth registries in Chile are universal, estimated to cover 99.9% of all births. Birth registries contain high quality records of birth weight (in grams), gestational length (in weeks), and size at birth (in cm), as well as information on the place of birth, and mother's and father's education and employment, along with other covariates.[8] Individuals are recorded using their national identity number, assigned at birth. Data on these births are merged (using a masked version of the national identity number) with the hospitalisation registry and the death registry, which cover all deaths and in-patient hospitalisations in the country. In total, 58,973 births are matched to a death record before the age of 1 year. And in total 83,841 births are observed to appear in the death registry at any point during this period.

For a number of mechanism tests, we consider hospitalisation records. Prior to 2001, individual identifiers are missing from a considerable proportion of micro-level registries of hospitalisations (inpatient records). Given this, we do not consider hospitalisation data prior to 2001, instead consistently working with subsets of birth cohorts for whom hospitalisation records are complete (for similar definitions see Menares and Muñoz (2025)). Thus, when we examine impacts of early-life health investments on later life health care usage, we work with age-specific outcomes. For example, when considering hospitalisation at age 1, we can examine this for cohorts whose hospitalisations are observed completely at this age, namely individuals born

[8] These data do not, however, contain information related to factors such as BMI, weight gain during pregnancy, smoking, or other behavioural measures.

from 2001–2017. And when considering hospitalisation at age 2, we work with the sample of births from 2000-2016, and so forth for other ages. Of all births, 2,924,795 are matched to at least one hospitalisation, and in total 5,656,409 hospitalisations are matched with births, implying that the average number of hospitalisations per matched birth is 1.93. These hospitalisations cover all inpatient care provided in the country, both in public hospitals and private clinics, and include information on both the reason for hospitalisation (recorded by standardised ICD-10 codes), as well as the duration of hospitalisation in nights.

Additional Data In a number of cases, we consider parental and family responses to child birth. In these cases, we generate information on changes following observed births, namely whether families go on to have additional births and if so the birth spacing in months, whether children are observed to be covered by private, rather than public, insurance or are treated in private, rather than public, hospitals, and whether parents are observed to leave or join the labour market, or move to a higher wage industry.[9] In each case, these measures are generated from administrative registers, and so are subset to individuals which appear in the relevant registers. For example, in the case of private insurance and hospitalisation, this measure is generated only for children who are observed to be hospitalised. Similarly, in the case of birth spacing and labour market changes, these measures are observed only for individuals whose parents go on to have a subsequent birth, as both of these measures are generated based off information contained in birth registries.

Sample Matches Sample matches are generated based on an individual's national identity number (the RUT) which is unique, ubiquitous, assigned at birth and used throughout life. In Appendix Table A1 we document the matches across registers. As noted above, not all births are matched with hospitalisations given that many individuals are never hospitalised, and not all hospitalisations are matched to births, as many hospitalisations occur to older individuals who were not born during the period of 1992–2018. Of particular interest is the cross within the birth registry, identifying the universe of all lineages where mothers were born from 1992 onwards. These 420,389 lineages by definition must only occur to mothers aged at most 26 years (mothers who were born in 1992 and had a birth in 2018). We document the full matrix of mother-child birth years in Appendix Table A2, which makes clear the composition of our intergenerational sample. Because for certain tests we consider individuals up to specific ages, we remove the small portion of individuals who have turned 26 years, as no observation in our sample has completed their 26th year of life. While this is a restricted sample, a feature of this sample is that it is somewhat similar in age range

[9] Birth registries contain information on each individual's occupation. We cross each individual's occupation with information on the average salary and average hours worked in each occupation by region from large household surveys conducted every 2-3 years (Chile's CASEN survey), thus allowing us to measure average conditions in the industry in which mothers and fathers work.

Table 1: Summary Statistics – All Births

	Obs.	Mean	Std. Dev.	Min.	Max.
Panel A: First Generation Births					
Gestation Weeks	6,574,154	38.64	1.82	15.00	44.00
$\geq$ 32 Gestation Weeks	6,574,154	0.99	0.10	0.00	1.00
Birth Weight in Grams	6,576,100	3323.61	539.29	90.00	6480.00
Birth Weight < 1,500	6,586,946	0.01	0.10	0.00	1.00
Birth Length in cm	6,570,257	49.37	2.58	9.00	59.00
Death Within 1st Year of Birth	6,586,889	0.01	0.09	0.00	1.00
Days Spent in Hospital by Year 1	838,941	10.31	23.36	1.00	365.00
Number of Admissions to Hospital by Year 1	4,221,971	0.28	0.72	0.00	81.00
Mother's Age	6,586,946	27.15	6.57	15.00	45.00
Mother's Education Years	6,572,902	11.43	3.25	0.00	21.00
Panel B: Second Generation Births					
Gestation Weeks	420,652	38.56	1.94	16.00	43.00
$\geq$ 32 Gestation Weeks	420,652	0.99	0.11	0.00	1.00
Birth Weight in Grams	420,631	3279.97	534.86	151.00	6375.00
Birth Weight < 1,500	421,050	0.01	0.10	0.00	1.00
Birth Length in cm	420,624	49.10	2.62	16.00	59.00
Death Within 1st Year of Birth	421,050	0.01	0.09	0.00	1.00
Mother's Age	421,050	19.48	2.62	15.00	25.00
Mother's Education Years	420,600	11.15	2.16	0.00	20.00

Notes: Summary statistics are displayed for all births occurring in the first generation (births from 1992-2018), as well as births matched to prior births (second generation births). The full sample consists of all births occurring between 1992 and 2018 in Chile from administrative data maintained by the Ministry of Health, and here we focus on mothers aged 15-45 at time of birth as used in principal estimation samples. Certain observations do not have information for certain variables based on exposure definitions: for example death within 1 year of birth is only defined for individuals aged at least 1 year, and number of hospitalisation days is only observed if individuals are hospitalised.

to that of East et al. (2023), who observe women up to the age of 28 when considering intergenerational links. This allows for some comparability of our results with theirs based upon women's age. In the case of Chile, in the period under study this is approximately the median age (refer to Appendix Figure A2). It is, however, important to note that this is a limitation in terms of data coverage, and in later years, we will observe younger women than our maximum age considered of 25. Figure A2d plots the age distribution of our intergenerational sample, and we discuss this limitation in Appendix A.1. In Section 5.2.3 we return to discuss the implications of this sample for the internal and external validity of our findings.

Summary Statistics Summary statistics for the matched microdata of the first and second generations are provided in Table 1. We observe average birth weights of around 3.3 kg among the full sample, and slightly lower when considering second generation births. These second generation births will necessarily be born to younger mothers (with an average age of 19.6 years, compared with 27.2 years in the full sample). A low proportion of births are VLBW (around 1% of the full sample), and on average most pregnancies are taken to full term (around 38.6 weeks in both the first and second generation sample). In Table A3 we document

identical summary statistics, but condition only on individuals born close to the 1,500 gram treatment cut-off. In this case we observe a much greater proportion of VLBW infants (given the sample definition), very premature infants, and deaths within 1 year of birth. In general, mothers in this sample are slightly older, at 28 years in the full sample.

4 Model and Methods

4.1 Justifying a Regression Discontinuity Design for Early Life Investments

As in Bharadwaj et al. (2013), consider the observed birth weight of individual i, denoted BW_i, which imperfectly proxies unobserved health at birth, H_i:

$$BW_i = H_i + \varepsilon_i.$$

Additionally, consider the inputs received by an infant at hospital, D_i, which depend, decreasingly, on health at birth, as well as discontinuous treatment assignment rules:

$$D_i = g(H_i) + \kappa \cdot \mathbb{1}\{BW_i < c\} + \nu_i,$$

specifically, here neonatal health treatments are shifted upwards by some amount κ when an individual is born with a birth weight below the cut-off c. In this setting, initial medical care D_i is correlated with unobserved health measures not captured by birth weight, implying that estimates of the impacts of early life health investments conditional on birth weight will be biased in standard models. This leads to the RDD:

$$y_i = f(BW_i - c) + \alpha \cdot \mathbb{1}(BW_i < c) + X_i\beta + \upsilon_i, \tag{1}$$

where $f(\cdot)$ captures local relationships between the running variable (birth weight) and outcomes of interest y_i, specifically allowing for split local-linear or higher order polynomials, X_i is a vector of covariates, and υ_i an unobservable error term. This design allows for the impact of medical treatments to be isolated from unobserved health stocks in the neighbourhood of c, given that crossing the threshold assignment causes a discrete jump in treatment $\Delta D_i = \kappa$, provided standard RD assumptions related to continuity of unobservables surrounding the cut-off point are met.

Estimation and Inference Procedures We estimate (1), with the baseline specification following quite standard procedures. Namely, our principal model consists of a split local-linear parametrisation for $f(BW_i <$

1500) within MSE-optimal bandwidths following Calonico et al. (2020), a triangular kernel, and robust bias-corrected inference of Calonico et al. (2014). We follow Bharadwaj et al. (2013) in defining specific details of the principal specification, diverging from this only where clearly-justified arguments exist, such as in the use of optimal bandwidth selection procedures, as these procedures have been largely developed following the publication of their paper. Thus, we control for covariates indicated by Bharadwaj et al. (2013) which are maternal characteristics (education, age, marital status), type of birth service (midwife or doctor), birth region, and year of birth. We additionally include a heaping control at 50 gram intervals, a point we turn to discuss below. While we aim to follow the models by Bharadwaj et al. (2013) as baseline specifications, we also document the stability of estimates to a range of alternative modelling considerations including alternative kernel weightings, polynomial orderings, and the use of donut RDDs (Barreca et al., 2011).

Validity of this RDD This RDD has been previously discussed and validated in a number of settings, and we conduct a suite of tests laid out in this literature. These are discussed at more length in Appendix B, and include tests for balance of predetermined observable factors around the cut-off, tests of heaping of the running variable following Almond et al. (2010); Bharadwaj et al. (2013); Barreca et al. (2011), tests for effects at placebo cut-offs, and tests for manipulation of the running variable. While our interest in this study is on the intergenerational impacts of early life interventions, we also document the initial relevance of these treatments in impacting infant mortality. These results are all discussed in Section 5.1.2 and Appendix B.

Principal Outcomes and Multiple Inference As we wish to consider how the impacts of early life health receipt accrue over life, and into future generations, we necessarily conduct multiple hypothesis tests considering multiple outcomes and a single treatment receipt. To avoid concerns that any findings will simply owe to inflated type I errors from repeated hypothesis testing, we proceed in two ways. Firstly, across all principal intergenerational measures we generate a single outcome index, following Anderson (2008). Secondly, within all classes of outcomes considered, we report both standard p-values, and q-sharpened p-values, which control for the false discovery rate.

4.2 What do reduced form estimates capture? A Conceptual Model

Understanding what α captures depends upon how posterior events and investments interact with D_i. Bharadwaj et al. (2013); Chyn et al. (2021) discuss this in a single generation setting (see Appendix C.1 where this is applied to our notation). We extend this into a multigenerational setting, where there are clear implications of selection into the second generation which may interact with initial treatment.

When we consider how early life treatments can have intergenerational consequences, this requires explicitly taking into account individual selection into appearing in the second generation. Specifically, we must consider that a first generation woman's fertility decisions may depend on her own initial inputs at birth, as well as the way these interact with future outcomes. Naturally, an individual will only appear in the second generation if a first generation woman has given birth. Thus, consider each woman's fertility decisions modelled at early ages (say her teenage years) and older ages (her 20s), which is described in terms of a series of latent variables as follows:

$$\text{(2)} \qquad \text{Early Fertility}_i^* = \gamma_1^E H_i + \gamma_2^E D_i + \gamma_3^E I^{post}(H_i, D_i) + \gamma_4^E B^{post}(H_i, D_i) + \upsilon^E$$

$$\text{(3)} \qquad \text{Later Fertility}_i^* = \gamma_1^L H_i + \gamma_2^L D_i + \gamma_3^L I^{post}(H_i, D_i) + \gamma_4^L B^{post}(H_i, D_i) + \upsilon^L.$$

Individuals for whom Early Fertility$_i^* > 0$ will have Early Fertility$_i = 1$, otherwise, Early Fertility$_i = 0$, with an identical crossing rule for Late Fertility$_i^*$. This is a simple model of fertility behaviour in which fertility at a given age is a function of an individual's health at birth H_i, treatment receipt at birth D_i, as well as total parental investment received after birth $I^{post}(\cdot)$, and individual behaviours after birth $B^{post}(\cdot)$. These latter two elements may both interact with initial health and medical treatment receipt.[10]

In turn, we may posit that birth weight of individual i's child (the second generation birth, who will be indexed j) can be described in the following way:

$$\text{(4)} \qquad BW_{ij} = H_{ij} + \phi H_i + \psi D_i + \varphi I^{post}(H_i, D_i) + \Psi B^{post}(H_i, D_i) + X_{it}\beta_t + \nu_{ij},$$

which is observed only if a generation 1 mother gives birth. Note that this function now depends on heritable factors, and so the mother's early life health, H_i, health interventions at birth D_i, as well as posterior investments in the mother, $I^{post}(\cdot)$, and behaviours of the mother, $B^{post}(\cdot)$, all may explain her child's birth weight. In (4) birth weight (BW) is used to fix ideas, but this is considered representative of the observed proxies of health at birth we discuss in Section 3.

Our interest in this paper is to determine the impact of intensive treatment at birth in the first generation on health outcomes of the following generation. Thus, our dependent variable of interest measures human capital at birth. To consider what an RD estimate captures where the outcome is birth weight of the second

[10] While this is a simplified model it provides a framework for understanding policy mechanisms, as discussed below. We document in Appendix C that it can be extended in multiple ways, which we will reference when introducing specific modelling elements. In Appendix C.2 we discuss modelling fertility at each age, and in Appendix C.3 note that health can be viewed as a dynamic stock without greatly altering implications for our setting.

generation, we can write the expected birth weight of the second generation *conditional* on the child appearing in the second generation in terms of the well-known Heckman (1974) selection equation. If we assume joint normality of each of υ^E and υ^L with ν_{ij} with arbitrary covariance terms ρ_E and ρ_L respectively, then, combining equations (2) and (3) with (4) we have:

$$\begin{aligned} E[BW_{ij}|\text{Early fertility}_i = 1] &= H_{ij} + \phi H_i + \psi D_i + \varphi I^{post}(\cdot) + \Psi B^{post}(\cdot) \\ &\quad + \rho_E \sigma \lambda_E \left[H_i, D_i, I^{post}(\cdot), B^{post}(\cdot)\right] \end{aligned} \tag{5}$$

$$\begin{aligned} E[BW_{ij}|\text{Later fertility}_i = 1] &= H_{ij} + \phi H_i + \psi D_i + \varphi I^{post}(\cdot) + \Psi B^{post}(\cdot) \\ &\quad + \rho_L \sigma \lambda_L \left[H_i, D_i, I^{post}(\cdot), B^{post}(\cdot)\right]. \end{aligned} \tag{6}$$

Here for simplicity we have written $I^{post}(H_i, D_i)$ as $I^{post}(\cdot)$, and similarly for B^{post}. A key point to note here is that selection is now captured by the (age-specific) inverse Mills Ratio λ, the standard deviation of ν_{ij}, and (age-specific) covariance term ρ. The assumption of joint normality is convenient in allowing a parametrisation of birth weight in terms of easily-understandable components in (5)-(6) although not required for our results below. We discuss a control function approach in Appendix C.4.

These selection equations (5)-(6) have clear implications for the RD estimate in the case where outcomes capture indicators of health at birth of the second generation. While the regression discontinuity design levies locally exogenous changes in treatment for first generation individuals, the total policy-relevant treatment effect on the second generation will consist of multiple posterior interactions with the policy. Specifically, noting from (1) that $\alpha \equiv \frac{\partial y}{\partial \mathbb{1}(BW_i<c)}$, and given the local smoothness assumption in RD which states that nothing varies around the cut-off except for treatment receipt, then $\alpha \equiv \frac{\partial y}{\partial D}\kappa$. Thus, based on the model laid-out above, the coefficient of interest when considering second-generation outcomes such as birth-weight can be written as:

$$\begin{aligned} \widehat{\alpha} \;=\; & \psi\kappa && \text{“Structural”} \\ & + \; \varphi \cdot \Delta I^{post}(c) + \Psi \cdot \Delta B^{post}(c) && \text{“Behavioural”} \\ & + \; \rho_E \sigma \Delta\lambda_E \left[c, \Delta I^{post}(c), \Delta B^{post}(c)\right] && \text{“Selection (Early Fertility)”} \\ & + \; \rho_L \sigma \Delta\lambda_L \left[c, \Delta I^{post}(c), \Delta B^{post}(c)\right] && \text{“Selection (Later Fertility)”} \end{aligned}$$

where $\Delta X(c)$ generically refers to any change in element X occurring when crossing the cut-off at c.

Based on this model, the reduced form impact of a mother i of child j crossing the 1,500 gram threshold is decomposed into five terms. First, $\psi \cdot \kappa$ denotes the direct intergenerational transmission of improved health at birth of the mother, on to her children (the ‘structural effect’ of the policy). Second and third,

$\varphi \cdot \Delta I_i^{post}(c)$ denotes the spillover on subsequent generations of parental investments in mother i which are sensitive to the policy and $\Psi \cdot \Delta B^{post}(c)$ refers to individual behaviours which are sensitive to policy receipt. This (grand-)parental investment channel will capture any changes in first generation parenting as a result of treatment-receipt which are then transmitted to the second generation, and the (parental-)behaviour channel captures any individual responses to the policy which may impact children. Finally, the fourth and fifth composite terms capture selection into the second generation, or a fertility channel of the policy which is allowed to be time-sensitive. This term, $\rho_E \sigma \Delta\lambda_E[c, \Delta I^{post}(c), \Delta B^{post}(c)]$ makes clear that this selection may operate in a number of ways. Given that the inverse Mills Ratio in equation (5) contains D_i as well as $I^{post}(D_i)$ and $B^{post}(D_i)$, selection may owe to direct policy impacts (e.g. healthier individuals as a result of intervention at birth may be more likely to take pregnancy to term), and/or selection may owe to changes in parental investments (e.g. compensating behaviour by individuals may increase the likelihood of fertility). In both cases, we are concerned only with the way which treatment receipt changes this selection, as if fertility is unchanged as a result of the policy $\Delta\lambda = 0$, and the fourth and fifth selection terms will be null. Note that, importantly, if fertility selection is relevant, the scale term ρ_E capturing unobserved correlations between birth weight and fertility likelihood will impact both the sign and magnitude of the selection term, underscoring the importance of understanding selection into fertility.

Implications This simple model has a number of clear implications, which we take forward in studying the mechanisms of effects documented in Section 5. One clear implication is that our RDD estimate includes multiple channels, such that the total effect of treatment need not be the direct effect of intergenerational transmission of health investments in observable stocks of health at birth (the 'structural' channel above). And the second implication is that the estimated quantity $\widehat{\alpha}$ can diverge from this structural effect if any of a number of channels are observed to be relevant.

The first channel is behavioural. If parental or individual behaviour changes when crossing the treatment threshold, this will potentially explain treatment effects provided that such behaviours affect health at birth. A second channel is selection into fertility. If we observe changes in rates of childbirth when crossing the treatment threshold, this will potentially explain treatment effects provided that unobserved elements of selection into treatment and birth weight are correlated. A third channel refers to timing of childbirth. Treatment receipt may shift individuals from later to earlier childbearing (or vice versa) even if it does not affect their total fertility, and this can potentially explain treatment effects provided that unobserved elements of selection into treatment and birth weight differ among younger and older women. In the results section we first present estimates of $\widehat{\alpha}$, before considering each of these theoretical channels as a guide for our analysis

of the mechanisms which explain $\widehat{\alpha}$. We finally comment on what one can infer about the direct structural channel when any of the other channels of intergenerational transmission are found to be present.

5 Results

5.1 Intergenerational Impacts of Health Intervention at Birth

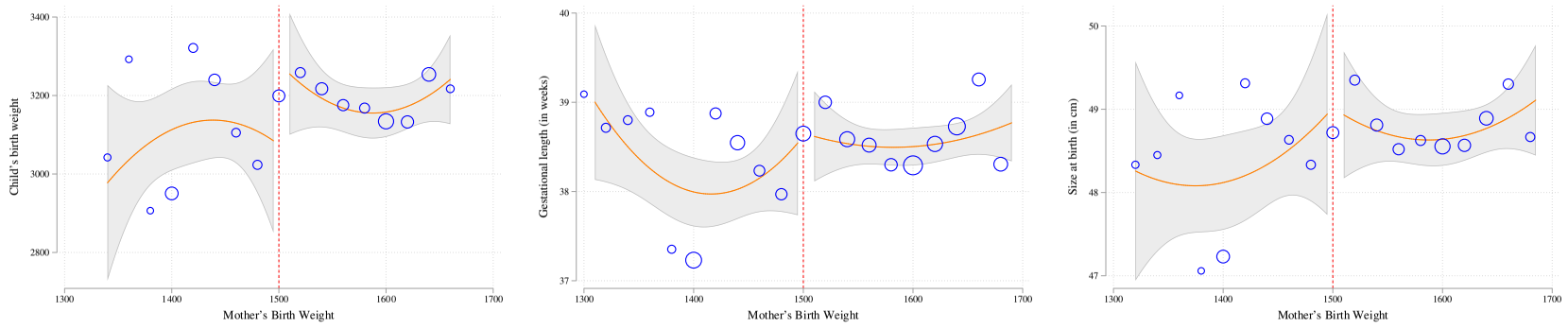
5.1.1 Principal Results

In Figure 2 we present visual RD plots, and in Table 2 we present formal tests for intergenerational transmission of intensive health treatments at birth. Coefficients capture the impact of a mother having received intensive treatment when she was born on her children's health when she gives birth many years later. Figure 2 plots average outcomes in 20 gram birth weight bins within Calonico et al.'s Mean Square Error-optimal bandwidth around the discontinuity, overlaid on quadratic fits and their confidence intervals. Table 2 presents robust, bias-corrected estimates based on local linear regressions with a triangular kernel. The stability of these estimates to alternative functional forms, bandwidth and modelling choices is discussed in Section 5.1.2 of the paper.

Across all outcomes considered, there is no evidence to suggest positive intergenerational gradients. For example, in the case of birth weight, there is a clear *upward* shift when crossing from below the cut-off (more treated) to above the cut-off (less treated) individuals in Figure 2a, and the point estimate in Table 2 shows a return of -217 grams on the next generation. This is a strong negative effect, significant at the 5% level if considering uncorrected p-values, or at the 10% level if consider the FDR corrected p-value (0.065). However, some of these estimates are imprecisely estimated. While there are a large number of intergenerational links covered in these 30 years of data (more than 400,000 for the 32+ week samples), when focusing on the optimal bandwidths this is reduced to a small number effective observations.[11] Nevertheless, despite noisy estimates, in all measures observed in Panel A point estimates are consistent with there actually being no, or negative transmission of early-life investments, rather than positive returns. As well as point estimates and standard errors (presented in parentheses in the table), we additionally present the p-value on a one sided test that corresponds to these early life investments having negative intergenerational impacts. Thus, a low p-value would point to evidence to reject the null that the effect of these investments are negative, and high p-values can be considered as providing very little evidence to reject this null. Across all outcomes considered, these p-values range from 0.716 to 0.984, suggesting very little evidence to reject the null of a negative effect in favour of the alternative of a positive impact.

[11] Power is a challenge in this setting. As we lay out in Appendix Figure B1.

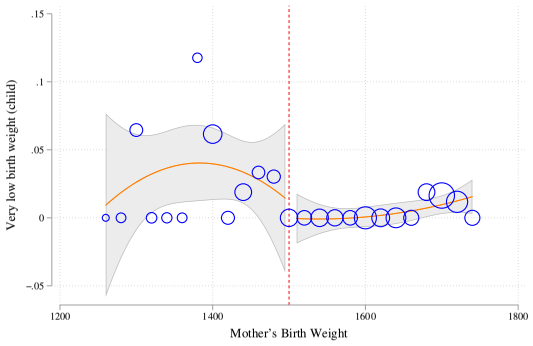
Figure 2: Descriptive Plots of Parental Policy Receipt and Child Health Measures



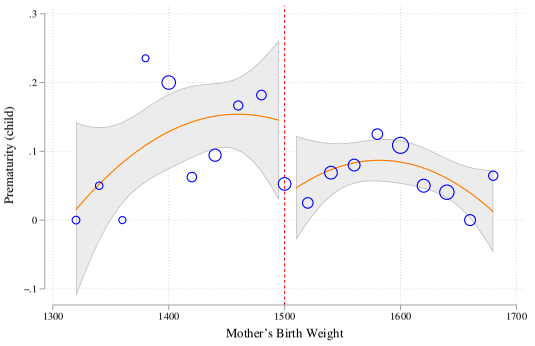
(a) Birth weight

(b) Gestational period

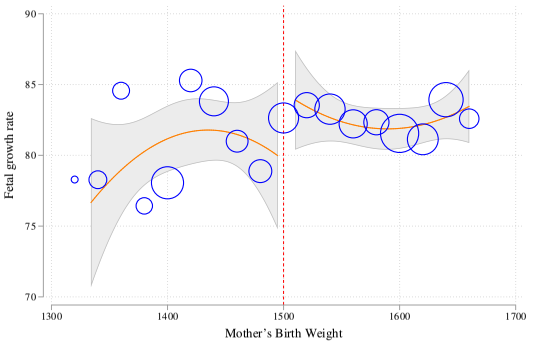
(c) Gestational length



(d) Very low birth weight



(e) Prematurity



(f) Fetal growth rate

Notes: Plots show separate quadratic fits estimated on each side of the 1,500 gram cut-off, in each case restricting attention to observations within the optimal bandwidth following Calonico et al. (2014). 95% confidence intervals of the quadratic fit are estimated, and circles represent average outcomes in 20 gram bins. The size of each point reflects the relative number of observations in each bin. Observations at 1,500 grams are plotted in binned averages, but are not used in estimating the quadratic fit.

Point estimates in the case of gestational weeks point to an insignificant reduction by around 0.6 weeks, and in the case of birth length an insignificant reduction of around 0.3 cm. Likewise, in the case of infant mortality of the second generation, we observe an insignificant effect, however the sign is consistent with individuals who receive intensive treatments going on to have children who are less likely to survive. In Panel B we consider a number of alternative derived measures based on variables recorded in the birth register. In the case of prematurity and low birth weight, these are critical measures of health stocks at points quite low in the distribution of birth weight and gestational weeks. In this case, at low points in the health distribution we find considerable negative returns to a mother's early life health receipt. In the case of both prematurity and low birth weight, we observe the individuals whose mothers just classified for treatment receipt are more than twice as likely to suffer from these conditions as individuals whose mothers were marginally above the treatment threshold. We similarly observe a statistically significant reduction in fetal growth rate (defined as weight divided by gestational weeks). The final outcome in Panel B is an index of health measures at birth which is used to reduce the dimension of the test to a single dimension, and avoid inflated type I error rates. Here, in line with results observed throughout the table, this child health index is observed to be considerably worse among children of parents who received early life treatment, at around 0.42 of a standard deviation *lower* on average.

The results from Table 2 suggest that while tests on mean outcomes point to negative but imprecise results, particular key points of the distribution of these outcomes do see substantial shifts as a result of the receipt of intensive early-life investments. This is particularly the case in low birth weight measures, where we observe that a mother's receipt of intensive investments at birth makes her far *more* likely to have low birth weight children. Given the high costs, both in terms of individual well-being as well as hospital-level costs for outcomes very low in the health distribution (Almond et al., 2010), we consider these distributional effects in a more flexible setting in Figure 3. In this case, each coefficient and confidence interval refers to the probability that a mother has a birth with health measures *below* particular cut-offs indicated on the horizontal axis. Each coefficient and CI is thus generated from its own RD estimation, and is interpreted as the impact of policy receipt on intergenerational transfer of poor health indicators.

Figure 3a shows clear negative impacts of maternal policy receipt on her child's health. In proportional terms, this is particularly relevant in the far left tail of the distribution of birth weight. For example, in the case of 1,500 grams, we estimate that a mother who receives intensive early life health treatment increases her probability of having a birth at less that 1,500 grams by 4.5pp, which is an effective tripling considering that the proportion of such births occurring to mothers in this bandwidth of interest are around 0.015. Similar

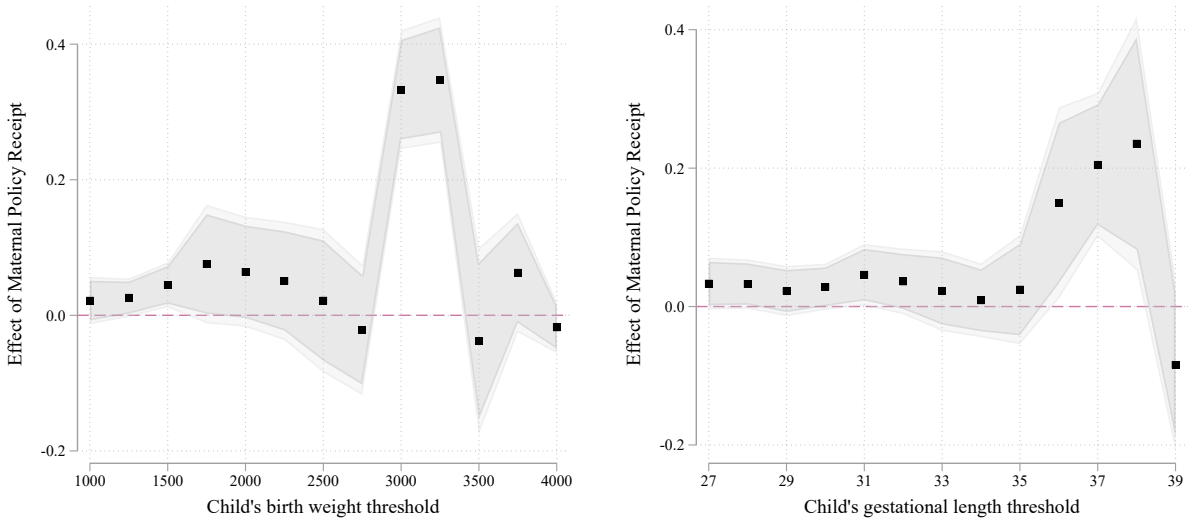
Table 2: Intensive Health Investments and Birth Outcomes of the Second Generation

Panel A: Baseline Variables	Gestation (weeks) (1)	Birth weight (grams) (2)	Birth length (cms) (3)	Infant Mortality (4)
Birth weight < 1,500	-0.563	-217.471**	-0.340	0.006
	(0.506)	(101.057)	(0.593)	(0.004)
	[0.867]	[0.984]	[0.716]	[0.940]
Mean of Dep. Var.	38.454	3160.307	48.612	0.007
Observations	416,913	416,892	416,886	417,304
Optimal Bandwidth	191.0	162.8	189.0	140.5
Effective Observations	863	763	836	669
Observations (left)	300	268	289	248
Observations (right)	563	495	547	421
q-sharpened p-value	0.154	0.065	0.204	0.079
Panel B: Transformed Measures	Prematurity (5)	Very low birth weight (6)	Fetal growth rate (7)	Anderson Index (8)
Birth weight < 1,500	0.150**	0.045***	-4.426**	-0.415*
	(0.071)	(0.017)	(2.156)	(0.236)
	[0.983]	[0.996]	[0.980]	[0.961]
Mean of Dep. Var.	0.085	0.013	81.919	-0.077
Observations	417,304	417,304	416,892	417,304
Optimal Bandwidth	182.7	242.3	167.5	227.5
Effective Observations	834	1,164	766	1,072
Observations (left)	289	347	271	338
Observations (right)	545	817	495	734
q-sharpened p-value	0.065	0.060	0.065	

Notes: Each column displays estimates of the change in the given dependent variable from above to below the 1,500 gram assignment threshold for mothers. In each case, local linear regression is used with a triangular kernel, calculating the MSE optimal bandwidth of Calonico et al. (2020). Robust bias corrected standard errors are reported in parentheses. Below standard errors, a one tailed t-test is calculated, which can be viewed as the support in favour of there actually being *positive* intergenerational transmission to the second generation. q-sharpened p-values refer to corrections conducted across the entire class of outcomes. * p<0.10; ** p<0.05; *** p<0.01.

such patterns are observed when considering weights up to 2,250 grams. It is important to note here that this implies that (costly) early-life interventions in generation 1 imply a much greater proportion of births having to receive such interventions in generation 2. In the case of gestational length documented in Figure 3b, similar such distributional impacts are seen, with consistently positive probabilities of observing short gestational lengths, significant in the case of 31 weeks (extreme prematurity), with particularly large impacts observed at 37 weeks (prematurity). Again, these results are consistent with *negative* intergenerational

Figure 3: Distributional Impacts of Early Life Health Interventions on Second Generation Health Stocks



(a) Birth weight

(b) Gestational length

Notes: Each point estimate (black square) and 90 and 95% confidence interval (dark and light shaded areas respectively) refer to RD estimates of the likelihood that a birth occurring to a treated girl born has health stocks (birth or gestational weight) below the cut-off indicated on the x-axis of each sub-plot. Each estimate is generated using the same sample and methods described in Notes to Table 2.

gradients of health intervention at birth, and certainly never suggestive of positive gradients as frequently observed in prior literature.

5.1.2 Robustness

The previous results are based on optimal RD procedures following standard choices, such as the use of bias correction, local linear estimation procedures, and weighting with a triangular kernel centred at the cut-off point. To avoid concerns related to specification search, we have precisely followed Bharadwaj et al. (2013) in the use of control variables. Nonetheless, we document the robustness of principal results to alternative specifications or empirical decisions. We summarise here tests presented in Appendix D which suggest these results are robust to alternative reasonable choices.

In Appendix Figures D1-D2 we document that results are broadly consistent across reasonable bandwidth choices, and evolve as expected as bandwidths grow considerably. We consider this in two ways: firstly (in Figure D1) we simply vary the bandwidth of data in which the RD model is fitted, using this same bandwidth for bias-correction, and secondly (in Figure D2) we increase the bandwidth, maintaining constant the ratio between the bandwidth used for estimation and bias-correction. Generally speaking, across intergenerational

outcomes, results are observed to be largest when focusing on a bandwidth more tightly bounded to the cut-off, and grow smaller only when the bandwidth is pushed up considerably, to as much as 300 grams. This value is well above optimal, and reassuring in that identification is local, and to the degree that wider bandwidths are used, more bias is expected.

More generally, we consider a range of alternative models varying, firstly the bandwidth selection and/or use of bias correction, and secondly the functional form of the running variable. These results are provided in Appendix Tables D1-D2. These consider a number of procedures for each intergenerational measure considered. First, we report robust-bias corrected results (replicating results from the main text). Second, we provide results using an alternative manner of selecting the optimal bandwidth which has been shown to be optimal to minimize rates of error in hypothesis testing (Calonico et al., 2020). Third, we present results using 'standard' RD models with conventional variance estimates, and fourth we present bias-corrected RD results, again using conventional variance estimates. Each of these results are presented for split linear and split quadratic polynomials to capture the relationship between outcomes and the running variable. Across models, results point to consistently negative intergenerational relationships (in the case of birth weight, prematurity, fetal growth rate, VLBW status and the Anderson (2008) index), or negative but insignificant results in the case of gestational weeks and birth length. One outcome (infant mortality) suggests noisier results, with signs flipping in certain cases, suggesting that this outcome should be considered with some caution, given its rarity in second generation births. Results are also robust to estimating a Donut RD with a range of donut hole radius values (Appendix Table D3), and do not in general appear to be driven by jumps owing to any heaping at rounded birth weights, with null effects largely observed when considering placebo designs at birth weights of 1250, 1750, 2000, 2250 and 2500 grams.

Finally, as we noted in Section 2, our estimation sample consists of all individuals born at weeks 32 and above, who are unambiguously exposed to the policy. We additionally present main results for the full sample of both individuals born at 32 weeks and above, as well as individuals born below 32 weeks. In general, we expect results may be slightly attenuated, given that it is less clear that these individuals are exposed to all aspects of the policy. These results are provided in Table D4 and Figures D3-D4. Results are quantitatively similar, though slightly dampened in certain outcomes.

5.2 Understanding channels of intergenerational effects

How can we rationalise the fact that we observe evidence which suggests *negative* transmission of early life health interventions across generations when the broad consensus from extant literature is that these

early life policies have broadly positive impacts on *first generation* individuals,[12] and positive shocks are generally observed to transmit across generations? It seems exceedingly unlikely that the direct effect of intergenerational transfers (the 'structural channel' laid out in Section 4.2) is negative, which points to some other relevant countervailing channel. The conceptual model laid out in Section 4.2 suggests three potential competing explanations, and we use this model to guide an analysis of these channels here.

5.2.1 Channel 1: Behavioural Responses to Early-life Treatment

A first channel which may explain these results owes to ways in which individuals and families react to treatment receipt. As highlighted in Section 4.2 parental investments may react to treatment receipt, or individual or family behaviour may otherwise change as a result of being exposed to treatment.

Compensatory or Reinforcing Behaviour A key channel which may explain null or negative intergenerational transmission of policy impacts relates to parental compensatory behaviour. As laid out in the model in Section 4.2, any change in parental behaviour as a *response* to treatment receipt at birth may act to counteract or reinforce any direct policy impacts. Specifically, a channel may exist in which parents whose children are born marginally above the 1,500 gram threshold and hence who are observed to be relatively worse off early in life, invest more heavily in these children, compensating initial disadvantage, and indeed fully closing the gap, explaining null or negative policy impacts in the long-run. This is something which can be empirically tested, if in place of examining the specific early life health measures in generation 2 births, we examine as outcomes in a RDD specification parental investment behaviours across the life of their children.

To do this we collect a number of measures of parental investments in their children, or parental behaviours which may impact children's outcomes or well being. These are classified in terms of health investments, future demographic decisions, and parental labour market sorting decisions. In the case of health investments, we examine whether, conditional on being hospitalised, children are covered by (more expensive) private insurance schemes, or treated in private hospitals, which implies higher average out of pocket spending (Crispi et al., 2020). In terms of demographic decisions, we examine whether treatment receipt impacts future parental fertility, either changing the number of future births or their spacing. In terms of labour market sorting, we test to see if *following on from their child birth*, mothers or fathers are observed to join or leave the labour market, to switch to a higher paying employment sector, or to switch to a less hour-intensive employment sector. Such labour market responses could potentially impact child well-being

[12] It is worth noting that in a cost-benefit sense, the positive first generation impacts in terms of lives saved far outweigh the costs of the programme, even when accounting for long term costs (see Appendix F).

on various margins: first an income margin if one or both parents opts to join to the labour market to afford greater investments in children, or second, a time investment if parents are observed to shift into less 'greedy' careers (as defined in Goldin (2021)), in favour of increasing (time) investments in children. Evidence of such labour movements and desires to seek careers which support both labour market and family investments are discussed, for example, in Goldin and Katz (2016).

Table 3: Parental Responses to Treatment Receipt

Panel A: Labour Market Responses

	Mothers				Fathers			
	Leaves Market	Joins Market	Expected Hour change	Expected Salary change	Leaves Market	Joins Market	Expected Hour change	Expected Salary change
Birth weight < 1,500	0.042	0.011	-0.275	77844.341**	-0.006	0.355***	-0.292	6803.405
	(0.056)	(0.038)	(0.292)	(31155.654)	(0.004)	(0.100)	(0.212)	(33673.293)
	[0.774]	[0.619]	[0.173]	[0.994]	[0.088]	[1.000]	[0.084]	[0.580]
Mean of Dep. Var.	0.230	0.237	40.528	541811.915	0.006	0.921	45.785	661338.944
Observations	366,044	669,779	558,787	554,751	856,752	47,842	1,044,260	1,018,479
Optimal Bandwidth	191.0	275.5	145.8	161.3	156.4	78.6	206.8	230.2
Effective Observations	834	2,097	877	1,001	1,438	38	2,396	2,703
Observations (left)	267	564	324	358	511	16	759	811
Observations (right)	567	1,533	553	643	927	22	1,637	1,892
q-sharpened p-value	0.548	0.724	0.548	0.046	0.362	0.003	0.362	0.724

Panel B: Parental Health Investments and Fertility Behaviour

	Private Insurance			Private Hospitalisation			Future	Birth
	2	4	6	2	4	6	Birth	Spacing
Birth weight < 1,500	0.045	-0.042	-0.032	0.060	-0.165*	0.079	0.018	13.372
	(0.058)	(0.050)	(0.072)	(0.066)	(0.093)	(0.129)	(0.020)	(10.245)
	[0.778]	[0.201]	[0.327]	[0.817]	[0.037]	[0.730]	[0.825]	[0.904]
Mean of Dep. Var.	0.194	0.221	0.227	0.289	0.333	0.359	0.301	196.483
Observations	197,745	153,707	107,870	197,745	153,707	107,870	3,556,247	1,363,510
Optimal Bandwidth	151.8	142.4	176.5	186.1	108.9	146.7	234.7	168.5
Effective Observations	772	485	403	929	349	336	14,132	2,874
Observations (left)	277	178	131	317	134	118	4,113	1,005
Observations (right)	495	307	272	612	215	218	10,019	1,869
q-sharpened p-value	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Notes: Each column displays estimates of the change in the given dependent variable from above to below the 1,500 gram assignment threshold for newborns. In each case, local linear regression is used with a triangular kernel, calculating the MSE optimal bandwidth of Calonico et al. (2020). Robust bias corrected standard errors clustered at the gram level are reported in parentheses. p-values for one-sided tests are shown in square (H_1: negative) brackets. Significance stars for two-sided test: * p<0.10; ** p<0.05; *** p<0.01.

We lay out tests of parental responses to treatment receipt in Table 3 Panel A (parental labour market responses) and Panel B (changes in health investments or future fertility behaviour). In Panel A, RD estimates suggest relatively little evidence of affected mothers changing intensive margin labour supply decisions (leaving or joining conditional on their previous labour market status), however we do observe evidence consistent with mothers of treated children moving into higher wage sectors or participating in sectors which

experienced higher wage growth. This is observed in column 4, with mothers of treated children being observed to change to sectors with mean monthly salaries which are approximately 80,000 CLP higher (around 80-85 USD based on exchange rates in 2025). In the case of fathers, we observe relatively little evidence of similar changes in labour market circumstance. The one exception to this is an increase in the likelihood that fathers of marginally treated children join the labour market conditional on previously having been out of the labour market, though we note that this effect is driven off a very small sample of fathers who previously had not participated in the labour market, and so is considerably underpowered. In general, this result suggesting larger labour market responses for mothers than for fathers in the face of child health shocks is consistent with findings in the broader literature, for example Eriksen et al. (2021) who find maternal shifts in labour market choices following negative child health shocks, and a broad literature, lead by Kleven et al. (2019), which documents relative inflexibility of father's labour supply to child birth, a phenomenon also noted in Chile (Berniell et al., 2021). While these parental labour market responses to early life investments are of interest in their own right, for our results here, if anything they suggest that the observed patterns cannot be explained by labour market results, as mothers of individuals born just below the 1,500 gram threshold are observed to move to industries with higher, rather than lower salaries, without being much greedier in terms of time demands.

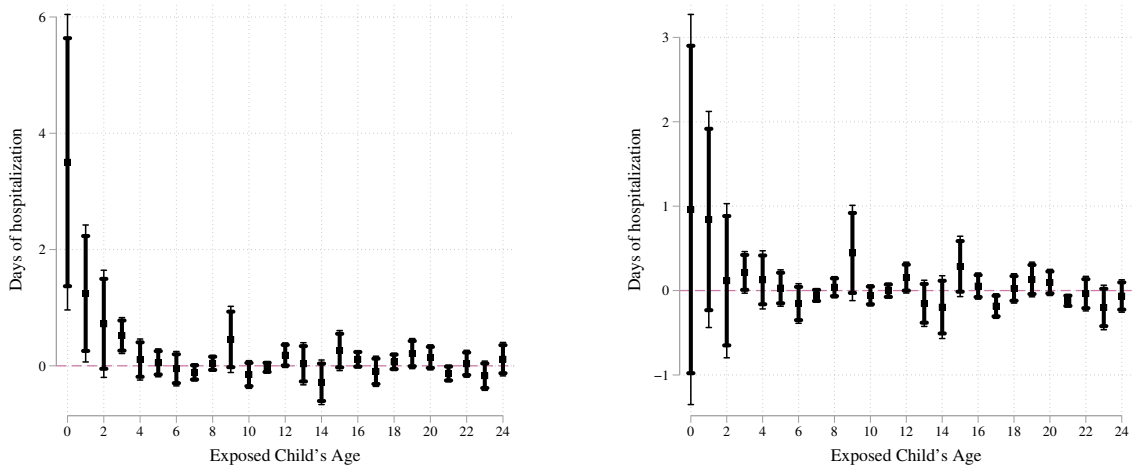
In the case of parental health investments and fertility responses (Table 3, Panel B), we observe relatively little evidence suggestive of consistent changes in the way which parents invest in their children's health care. This coheres with evidence presented by Bharadwaj et al. (2013) who found relatively little evidence of changes in educational investments by parents of marginally treated versus marginally untreated children. Early in life, we observe that conditional on hospitalisation, rates of private hospitalisation and private insurance coverage did not significantly differ across the 1,500 gram threshold. Similarly, we observe no evidence to suggest that parents of treated individuals were more likely to go on to have another birth, or change the timing of future births. In fact, both birth timing and the number of future births following a very low birth weight baby are quite similar to outcomes in the general population (Appendix Figure A3).

In general, these results suggest that (first generation) parental responses can explain relatively little of the observed negative intergenerational impacts to children of the second generation. Had we observed clear evidence of compensating investments, where individuals just to the right of the cut-off received additional investments or otherwise more positive home environments, this may have suggested that initial favourable medical treatments of treated individuals were overwhelmed by later favourable investments in untreated individuals. If anything, we observe that the reverse may be true, given the relatively better labour market

trajectories for mothers of treated children.

Behavioural Responses to the Policy: Interaction with the Health System One potential explanation for these results is that a mother's stock of health at birth is passed on to her baby. Specifically, given the positive relationship between maternal health stocks and child health at birth (Lassi et al., 2013; Currie and Cole, 1993), for a health channel to explain *negative* intergenerational transmission, we would require that mothers who were marginally treated by the policy at birth be *less* healthy than mothers who were marginally untreated. While unlikely, such a phenomena could occur, if, for example, individuals who were less treated at birth were more likely to be hospitalised in early life, and ended up accruing more positive health stocks by maturity. Thus, the key consideration for this mechanism is whether health stocks appear to be different, and in particular, whether there is evidence of compensatory investments in health during an individual's fertile period.

Figure 4: Long-Term Health Stocks and Early Life Interventions



(a) Hospital days **(b)** Hospital days, 'chronic' causes

Notes: Each point estimate and confidence interval refer to the impacts of early life medical investment on an individual's days of hospitalisation (panel (a)), and days of hospitalisation related to chronic conditions (panel (b)). Thicker black error bars present 90% CIs, while thinner error bars report 95% CIs. Days of hospitalisation (in both cases) are measured as totals for all individuals who have reached the age indicated, and take the value of 0 if the individual is not hospitalised this year, or otherwise a positive integer reporting the total number of days spent in hospital. All estimates follow the procedures laid out in Section 4, and report RBC estimates using a local linear regression with a triangular kernel in the MSE optimal bandwidth.

In Figure 4 we examine the number of days an individual is hospitalised by year, estimated following the RD models laid out above at each age from 0 up until 24. Figure 4a estimates the impact on the total number of days that affected individuals spend in the hospital for all causes, and panel (b) estimates the impact of

early life medical treatment on hospital days which are classified as chronic.[13] In both panels it is apparent that there are relatively small or non-existent impacts on the long term usage of health care. Both when considering all hospital days, and hospital days for chronic causes, after the first few years of life there is little evidence of an enduring effect on hospitalisations.

We *do* observe a clear policy impact on hospitalisations early in life, estimating an increase of around 4 days in the year which birth occurs, 0.5-1.5 days per year up to year 3, and then smaller and non-significant impacts there-after. These effects are in line with those documented in Bharadwaj et al. (2013), reaffirming the power of the policy in terms of health investments. However, the lack of clear later life results suggests that a role of direct transmission of health stocks from mother to children may be limited, given that we do not observe that individuals who marginally qualified for treatment have considerably different health stocks at maturity, at least if health stocks are proxied by hospitalisations.

Behavioural Responses to Treatment: Education, Partnership, Family Spillovers and Migration There are a range of alternative possible explanations which we are not simple to test with the available data, but which can be illuminated by examining the available literature or aggregate statistics. This includes educational attainment and school peer interactions or relationship formation more generally, within-family spillovers, and selective out-migration.

Education Impacts of early life medical receipt on education have been documented in a number of settings (Bharadwaj et al., 2013; Daysal et al., 2022), and so may partially explain observed results. Bharadwaj et al. (2013) documents that educational outcomes up to grade 8 improved on average among individuals who marginally received treatment, while Daysal et al. (2022) find improvements in education in ninth grade test scores, though no significant effect on the likelihood to enrol in higher education. While our data is fully matched within the health system, we are not able to observe educational attainment of individuals who do not give birth, and so cannot directly test whether education is observed to increase in this setting and age group more generally. However we are able to observe education at the age when mother's give birth, and in Table D6 document that results are found to hold when conditioning on both mothers and grandmother's education at birth, as well as when only focusing on individuals from highly educated families (as proxied by grandmother's education, given that young mothers are potentially still completing their education). We find results are broadly consistent when controlling for education, and when conditioning on high education

[13] Chronic hospitalisations are classified using the Chronic Condition Indicator (CCI) developed by the Healthcare Cost and Utilization Project (HCUP).

(and indeed, are slightly larger in this group). Because education is, if anything, likely to be higher among individuals marginally below the 1,500 gram threshold, it seems unlikely that educational attainment itself can explain worse outcomes among individuals who received treatment. However, we return to explore this point in more detail in Sections 5.2.2-5.2.3, where we consider whether educational attainment may interact with any selection into child birth, or timing of child birth.

Partnership A second explanation relates to individual interaction with treatment and the marriage market. One possibility is that individuals who are exposed to early life treatment form better partnerships, with impacts on support during pregnancy and outcomes at birth. Once again, as in the case with education, this seems unlikely to drive results in a direct fashion, and in Table D6 we show that results are robust to controlling for a proxy of partner presence as well as considering only individuals with observed partners. As in the case with education, we are not able to observe partnership characteristics of women who do not give birth, and so return to discuss this point in Section 5.2.2-5.2.3 when considering whether this interacts with selection into second generation birth or birth timing.

Alternative Behavioural Explanations: Within-family spillovers or Migration There are precedents suggesting that early life health interventions at this margin can result in positive within-family spillovers, improving mother's mental health as well as sibling outcomes (Daysal et al., 2022). While our data does not allow us to consider such measures in this context, for this to explain a *negative* intergenerational health spillover, these positive family spillovers such as improved maternal mental health would need to map negatively into health of the following generation, which seems unlikely. One final explanation related to selection owes to migration out of the country. Such migration would need to be selective on one side of the treatment threshold. This explanation also seems unlikely given that emigration rates in Chile are very low, at 0.35 per 1,000 individuals, and generally concentrated at working ages, particularly from 35 onward (Stefoni, 2011).

5.2.2 Channel 2: Selection Into Fertility

A second theoretical channel which may explain negative intergenerational effects is treatment-mediated *selection into fertility*. If as a result of treatment individuals give birth when they would not have given birth without treatment, this could potentially explain the results documented in Section 5.1. Indeed, it is also true that selection could originate even at the time of birth of the first generation given that the treatment impacts survival of (potential) parents when they are born. While here our focus is on selective fertility, we discuss

briefly first that survival selection when originally receiving treatment is unlikely to drive our results.

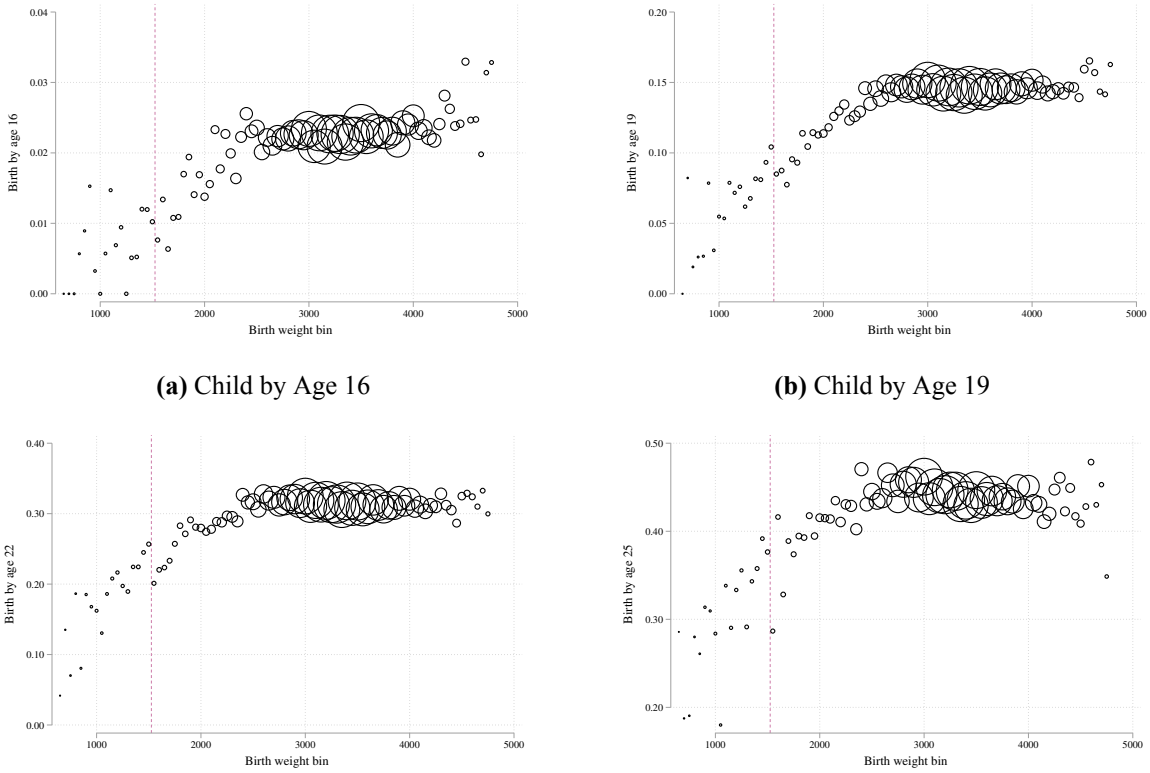
(A) Survival selection in generation 1 While the main analysis focuses on selection into second-generation births through differential fertility, a related concern is that the policy may also have induced selection through survival at birth in the first generation, given evidence that intensive neonatal treatment increased survival among infants with poorer initial health stocks. To assess whether such selective survival could account for the negative intergenerational effects we observe, we conduct counterfactual exercises that impute the fertility and health outcomes of individuals who would have survived absent the policy. This is discussed at more length in Appendix E.1 (Table E1 and Figure E1). Across a wide range of scenarios—varying assumed health at birth across the observed distribution and imputing fertility profiles based on birth-weight-specific averages—selective survival can attenuate the magnitude of estimates but is generally insufficient to reverse their sign. Even under extreme assumptions, such as highly fertile counterfactual individuals giving birth to children at very low health percentiles, negative intergenerational effects persist for most outcomes, including birth weight, gestational length, and fetal growth. Only for size at birth do more plausible counterfactuals partially offset the estimates. Overall, these results suggest that selective survival at birth in the first generation alone is not sufficient to explain away the observed negative intergenerational transmission across health dimensions.

(B) Selection into fertility later in life Figure 5 presents descriptive plots of the likelihood that a woman gives birth by specific ages, graphing probabilities by an individual's own weight at birth. These plots are based on all women exposed to the possibility of giving birth—that is individuals who have reached the age under consideration in each plot, and hence are potential 'second generation' mothers. Plotted values represent the actual proportion of these women who actually do become mothers. Each point refers to the average in 50 gram bins, with the size of the point representing the number of individuals on which the average is based. Any 'selection' of interest would be apparent by changes in rates of birth just at the point where treatment targeting ends.

Prior to considering selection *per se*, it is immediately notable and noteworthy that there is a steep gradient in the likelihood of giving birth by each of the ages documented in Figure 5 at lower points of the birth weight threshold. For each of the ages documented, there is a steep gradient in rates of birth up to around 2,500 grams.[14] For example, in the case of births by the age of 19, individuals who survive to age

[14]For ease of visualization, only 4 plots are displayed in the main text. In a previous working paper version of this study (Clarke et al., 2022) we have documented patterns for each age.

Figure 5: Fertility and Stocks of Health at Birth



(a) Child by Age 16

(b) Child by Age 19

(c) Child by Age 22

(d) Child by Age 25

Notes: Each figure plots the likelihood that a women born at a particular birth weight goes on to give birth by the age indicated in the plot caption. Points represent average proportions in 50 gram bins based on the women's weight when she was born. Averages are calculated from the full sample of women observed in the birth register, who have reached the age indicated in each plot. The size of each point refers to the relative number of women in this birth weight bin. The red dashed line indicates the 1,500 gram VLBW threshold used to assign discontinuous medical intervention at birth.

19 and were born below 1,000 grams have less than a 10% chance of having given birth, rising to around 12% among individuals between 1,500-2,000 grams, before levelling off at 15% above around 2,500 grams. While points low in the birth weight distribution are based on few observations, the regularity of this pattern, both within and across age groups is clear, with this gradient consistently being observed, and consistently flattening out from above around 2,500 grams.

As far as we are aware, this stylised fact has not been previously documented in the economic literature, and the few medical studies which have documented correlations between birth weight and fertility are observational and based on relatively small convenience samples (Vikström et al., 2014; van der Pal et al., 2021; van Gendt et al., 2015), or indeed consider other birth outcomes such as preterm birth (Swamy et al., 2008). These results suggest considerable returns to birth weight in ways not previously considered in the economic literature. It also interacts with a broader literature on labour market returns to health at birth, and

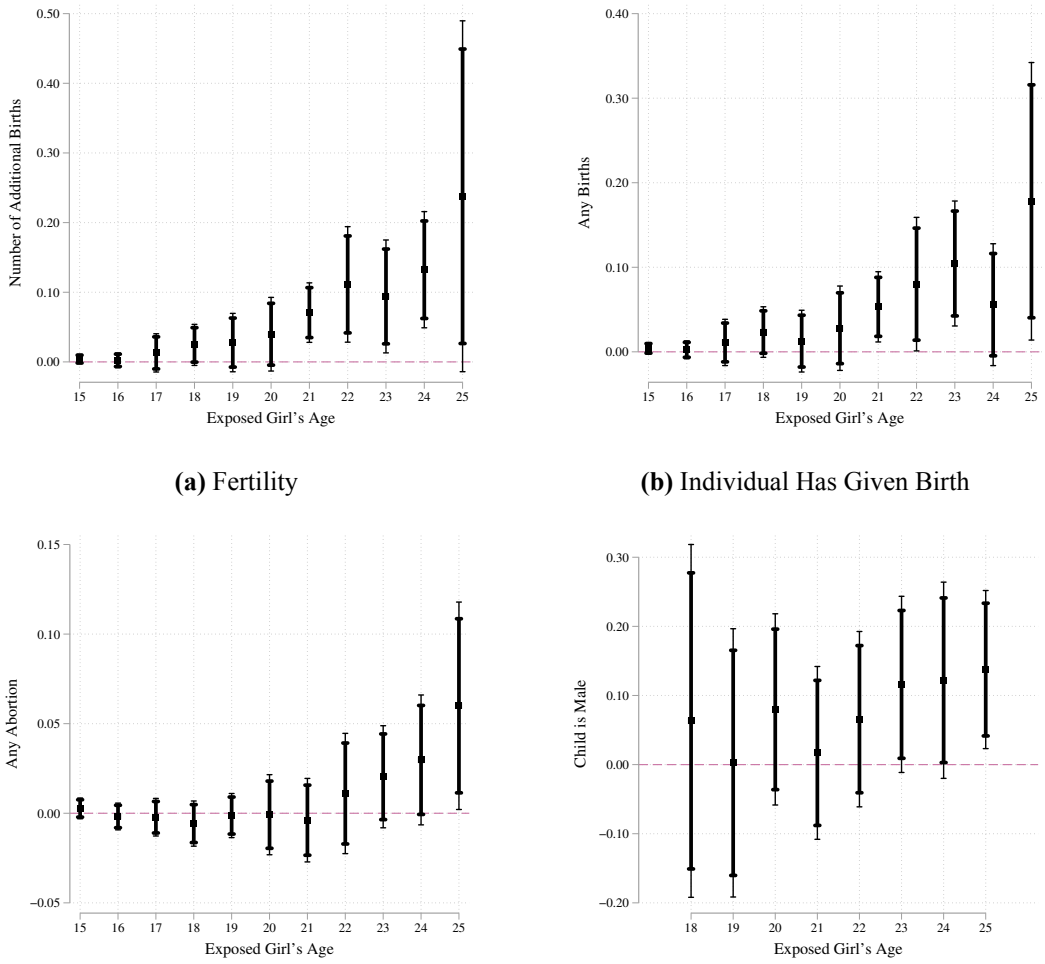
women's labour market returns in particular. If higher birth weight individuals have higher educational and labour market returns (Behrman and Rosenzweig, 2004; Bharadwaj et al., 2018), and at the same time birth weight is positively correlated with fertility, then negative links between fertility and labour market outcomes (Adda et al., 2017; Bloom et al., 2009) may partially obscure the full human capital returns to birth weight.

In Figure 5 it is additionally notable to the plain eye that there is an important discontinuity in rates of birth which are observed around the 1,500 gram cut-off. Even in using quite crude 50 gram bins in the spirit of capturing descriptive patterns, in each case clear sharp declines are observed in rates of birth when moving from just below 1,500 grams (marginally treated individuals) to just above 1,500 grams (marginally untreated individuals). For example, in Figure 5c, around 25 percent of individuals are observed to have a birth before the age of 22 in the 50 gram bin just below 1,500 grams, with this proportion falling to around 20 percent, or declining by approximately one quarter, when moving to the bin just above 1,500 grams. These values are identified formally using identical RDD methods as used throughout the paper in Figure 6. Here age-specific estimates are presented, estimated following equation 1, where the outcome considered is the total number of births a (potential) second generation mother has by age x (panel (a)), or if she has had a birth by age x (panel (b)). We observe large, and generally significant, impacts across all ages considered. By the age of 22, estimated impacts described in Figure 6a show that the number of births for individuals receiving the policy has increased by 0.1, against a base of approximately 0.28 births in individuals in the estimation bandwidth. Similarly large estimates are observed at each age above 20, growing to 0.2 additional births by the age of 25, versus a mean of around 0.48. These values are age-specific and grow by age, although if we estimate a model pooling across all ages, this suggests that fertility increases by a statistically significant 0.048 births, versus a mean of 0.214 births (Table E2).

This result suggests that policy receipt has a clear long-run impact on treated girls. It makes them considerably more likely to give birth, as much as a quarter of a century after the initial policy receipt. This also provides a key potential explanation of the unexpected *negative* transmission of health at birth into the following generation. The policy may have a positive effect on fertility by rescuing marginal births of treated individuals. Individuals who in the absence of the policy would not have conceived or not have given live birth to a baby, do give birth to a baby when receiving intensive treatment at birth. This finding is consistent with the policy saving (second generation) babies which are marginally weaker, providing one explanation of the negative impacts observed in Section 5.1 of this paper. We return to discuss evidence in favour of this interpretation in Section 5.3.

Fertility selection is thus a key mechanism which can potentially explain the negative transmission across

Figure 6: Impacts of Early Life Health Interventions on Fertility and Spontaneous Abortions



(a) Fertility

(b) Individual Has Given Birth

(c) Abortions

(d) Male Child

Notes: Each point estimate and confidence interval refer to the impacts of treatment on the number of births an individual has had by each age (panel (a)), the probability an individual has had a birth by each age (panel(b)), the number of abortions observed in hospitalisation data (panel (c)), and the likelihood that an observed birth is a male (panel (d)). Estimates in panel (a)-(c) are defined for all second generation individuals surviving to each indicated age, and so each estimation sample consists of all individuals. Estimates in panel (d) are limited to individuals who have had a birth by the age indicated on the x-axis, and so are limited to births occurring at (at least) age 18, as otherwise samples are too small to reliably estimate RDD models. Error bars 90% and 95% CIs. All estimates follow the procedures laid out in Section 4. Results for all women including those born below 32 weeks are provided in Figure D5.

generations. We briefly consider what explains this fertility result (the 'mechanism of the mechanism'), prior to turning to potential alternative selection channels. Higher rates of childbirth following a shock such as policy receipt may occur in a number of ways. Firstly, it may be the case that individuals are equally likely to conceive, but just less likely to take births to term when they do not receive the intensive early life medical investment. Or secondly, it may be that the early life medical intervention provides individuals greater social or reproductive resources to conceive. Due to the nature of administrative records and high rates of

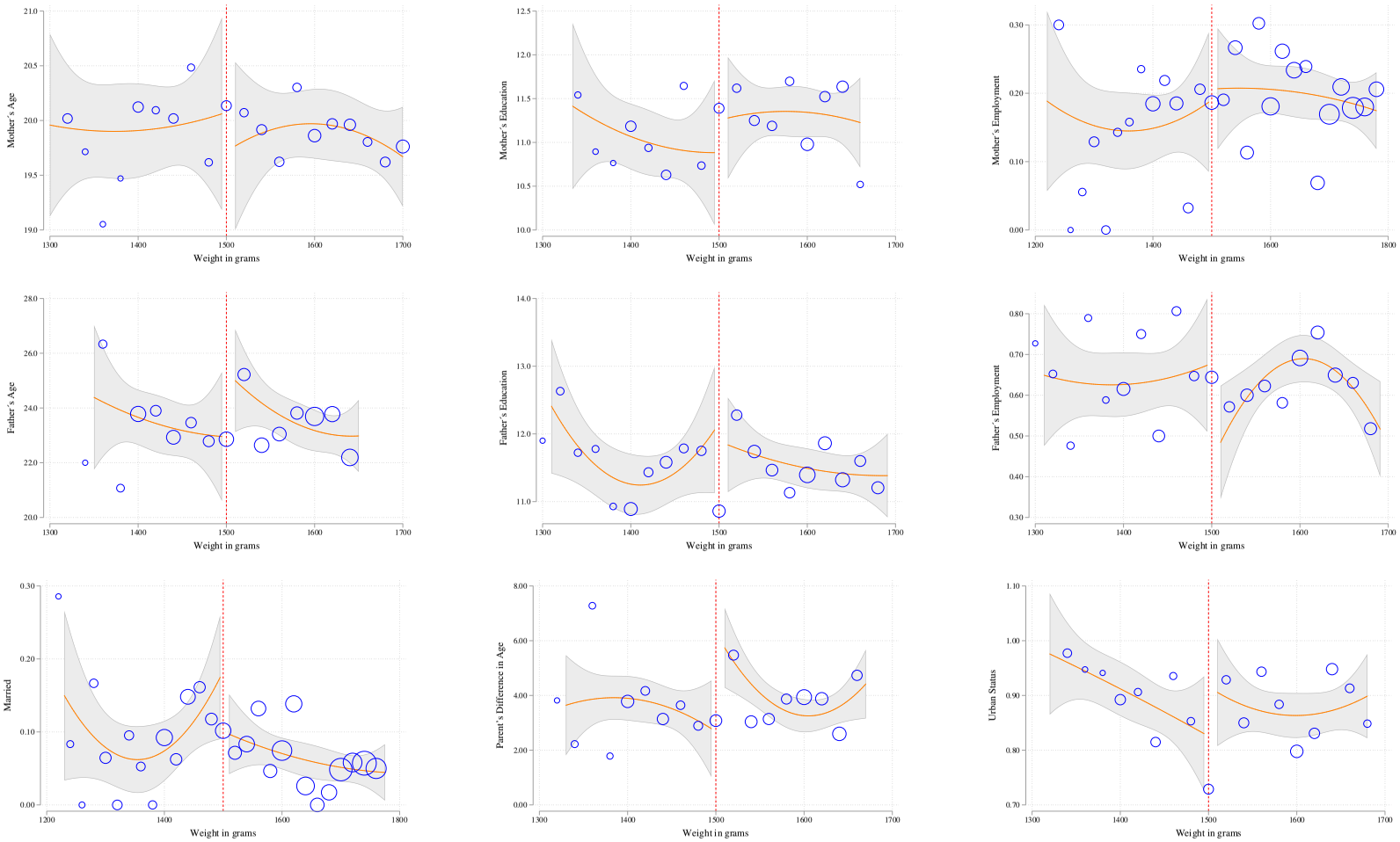
miscarriage, we cannot observe all conceptions occurring around the cut-off, but rather only actual birth rates. As a proxy of miscarriage, or births not taken to term after conception, we can observe all spontaneous abortions which result in hospitalisations.[15] This is likely a considerable lower bound of all conceptions not taken term. We examine the impact of policy receipt on the likelihood of suffering a spontaneous abortion in Figure 6c. Analysis is conducted identically to the analysis of fertility. Here it is appears that based on *this* proxy, higher rates of birth among treated individuals do not owe to a greater likelihood of taking births to term, but rather, more likely due to a greater propensity to conceive.

However, an alternative and potentially more sensitive proxy of selective survival in utero is the sex ratio of births. It is well known that male fetuses are less likely to survive to birth, and are more sensitive to shocks in utero (see, *e.g.* Valente, 2015). Thus, an improvement in rates of survival of males at birth is evidence of improvements in rates of survival among relatively weaker babies (Trivers and Willard, 1973). We replicate our analysis from panels (a) to (c) with the likelihood that a child is male in panel (d). Based on this measure, we *do* see evidence in favour of fertility results being driven by selective survival of weaker fetuses in exposed mothers. In Figure 6d we observe substantial improvements in rates of birth of male children on the policy-exposed side of the RD threshold, and this becomes particularly noteworthy at ages where fertility increases most. These effects are large, suggesting around a 10pp increase in the likelihood of male children being observed among live births.[16] This suggests the observed results can, at least in part, be ascribed to birth selection occurring after conception, and potentially relatively early in gestation given a lack of clear impacts on spontaneous abortion. It is also important to note that this explanation coheres with the distributional results we observed in Section 5.1. If treatment receipt saves second generation births which would not have been taken to term without the policy, we may expect increases in births with very poor health outcomes, which is what we did in fact observe with indicators of very low birth weight.

[15] These are observed in inpatient records, and are inferred from ICD-10 codes which are recorded as the reason for treatment. A full tabular classification is provided in an earlier version of this working paper (Clarke et al., 2022).

[16] This is apparent even among descriptive statistics, with the likelihood of a birth being male in the 250 grams below the threshold being 0.47, compared with 0.52 in the 250 grams above the threshold.

Figure 7: Is There Policy-Driven Selection into Childbirth?



Notes: Each plot documents characteristics of mothers, fathers, or births of second generation births surrounding the 1,500 gram treatment application threshold of first generation births. A quadratic polynomial fit is graphed on either side of the cut-off, and points reflect average outcomes in 20 gram birth weight bins, with relative sizes reflecting the number of individuals in each birth weight bin. Each plot is displayed within Calonico et al. (2020)'s MSE optimal bandwidths. Formal tests of discontinuities at 1,500 grams are provided in Appendix Table E3.

This mechanism is also able to rationalise the effect sizes documented when considering second generation outcomes at birth. Consider for example the impact on rates of VLBW in Table 2, which is an increase of 4.5pp, compared to a mean rate of 1.5pp among unexposed individuals. In essence, what this means is that the proportion of VLBW births jumps from 1.5% to 6% when crossing the treatment threshold. Note also that we find that intensive medical treatment increases births by around 20.4% (0.048/0.235 from Table E2), and these infra-marginal births are those births which would have not survived without treatment receipt. It is most likely that a very high proportion of babies that do not survive are very low birth weight (and generally have very poor health stocks). However, even if we take a reasonably conservative view and assume that only around one quarter of these births which did not survive without policy intervention would have been very low birth weight, this can justify our effect.[17] The fact that we find a quite large effect on fertility, and that infra-marginal births will naturally be those with poor health stocks, suggests that our effect sizes are credible, and smaller effect sizes are unlikely unless infra-marginal births have unexpectedly high measures of health at birth.

In Figure 7, we examine whether this fertility selection is a generic result in the population, or something which is driven by particular groups. Formal RD estimates which correspond to each plot in Figure 7 are provided in Appendix Table E3. To test for selective fertility, we examine characteristics of mothers and fathers who go on to give birth in generation 2 around the 1,500 gram birth weight threshold. This figure is thus the intergenerational analogue of typical balance tests conducted in RDD to examine whether observations on each side of the cut-off are balanced, or are selected in some way. However, here, rather than acting as an identification check, these allow us to test for selective entry into the second generation of mothers. Across 9 outcomes observed in administrative data (mother's and father's education, age and employment, marital status, and whether births are multiple), we observe some evidence consistent with policy-driven fertility changes being more prevalent among certain groups. For example, we observe weak evidence to suggest that less educated women are slightly over-represented among treated rather than control individuals, which would be consistent with marginal births being more likely to survive among less educated women when they receive the treatment, compared with when they do not. We observe more clear evidence to suggest that treated individuals have *more* preferable partner matches when partners are observed: fathers are on average closer in ages to mothers, more likely to be employed, and mothers and fathers are more likely to be married. Such a result may be consistent with treated individuals more generally forming better partnership matches as a result of treatment receipt. In the following sub-section we consider some suggestive tests related to

[17] This is calculated by dividing the 347 births in the below threshold (treated group) in Table 2 into infra-marginal births (20.4%, or 71 births) and always-survivors (79.6%, or 276 births). Given that on average 6% of surviving births are VLBW, we can calculate the rate of VLBW among infra-marginal births as: $(0.06 \times 347 - 276 \times 0.015)/71 = 0.23$.

whether compositional changes in parents can explain our observed results.

5.2.3 Channel 3: Fertility Timing

The relevance of selection into fertility implies that it is important to consider birth timing. This is important for two reasons. A first is related to the internal validity of these estimates. As we consider mothers up to the age of (at most) 25, if treatment receipt makes mothers more likely to give birth early in their life, but then they do not give birth later, our estimates may simply be capturing a mechanical change in health at birth given that (over the age horizon we consider), children born to younger women have poorer stocks of health.[18] We document this fact in Figure E2 observing that descriptively, there is a steep gradient in health at birth as mothers age between 15 years until around the mid-20s, at which point the gradient is reversed. What's more, this pattern is observed to hold even in mother fixed effect models, where identification is drawn off of variation in birth timing within individuals. A second reason is related to the external validity of these estimates. In a similar vein to the argument laid out above, if treatment receipt is simply causing individuals to shift births from later in life to earlier in life, and if this is specifically the case for individuals who are negatively selected on health, the estimates we report in this paper may be entirely reversed as women age. We focus here on these questions, most specifically those of internal validity where we observe relevant information, though comment briefly on questions of external validity.

On the question of internal validity, first we can ask if we actually *do* observe patterns consistent with births being shifted from earlier later to earlier ages. Descriptively, this does not appear to be the case. In Figure E3, we see that rates of birth generally open up from a young age in treated groups, and then do not show signs of later closing back up.[19] More formally, we test whether we observe differences in the age composition of births as a result of treatment receipt in a series of balance tests. We consider this in Table E4, which consists of a series of regressions of age indicators on treatment receipt within the bandwidth used in Section 5.1. When considering the full sample, (column 1), we do not observe evidence to suggest substantial shifts in the age composition of births above and below the cut-off. This suggests that the selection into fertility is constant across ages rather than shifting individuals from older to younger ages.

It may be the fact that while we do not observe shifts in the total number of births, the composition of who gives birth when changes in a way which is concerning for internal validity. In examining balance tests

[18] While results are necessarily noisier when stratifying by age, suggestive results presented in Figure D6 point to effects that are large both at younger and older ages among the groups considered, potentially limiting these concerns (at least within the age groups studied). If we are instead concerned that there may be imbalance over time with later birth cohorts observed until younger ages, we can rule out this concern by considering effects with a balanced sample of birth cohorts, as documented in Table D5.

[19] These patterns are clearly different to those observed if we focus on other (arbitrary) cut-offs such as above and below median birth weights (Figure E4).

Table 4: The Effect of Compositional Change on Health at Birth

	Gestation Length (1)	Birth weight (2)	Size (3)	Infant Mortality (4)	Premature (5)	VLBW (6)	Fetal Growth (7)
Panel A: Birth Timing Only							
Compositional Effect	0.00068	1.851	0.00332	-0.00001	-0.000194	-0.00001	0.0474
Proportional Effect	-0.121	-0.851	-0.978	-1.194	-0.130	-0.130	-1.071
Panel B: Birth Timing & Education							
Compositional Effect	0.00536	8.386	0.0177	-0.00031	-0.00078	-0.00022	0.207
Proportional Effect	-0.953	-3.856	-5.213	-4.816	-0.522	-0.492	-4.677
Panel C: Birth Timing & Partnership							
Compositional Effect	0.00196	-1.840	-0.00124	0.00001	-0.00015	-0.00001	-0.0509
Proportional Effect	-0.348	0.846	0.365	0.916	-0.0990	-0.0147	1.150
Panel D: All							
Compositional Effect	0.00673	6.544	0.0152	-0.0002	-0.00067	-0.00014	0.154
Proportional Effect	-1.196	-3.009	-4.477	-3.148	-0.449	-0.301	-3.479

Notes: Compositional effects are estimated as the expected change in outcome if instead of the composition of births among treated individuals, treated births had followed the composition observed among untreated individuals based on estimated age effects from mother fixed effect models documented in Figures E2-E5. Observations used in each column refer to the optimal bandwidth from Table 2 Proportional effect refers to the % of the estimated treatment effect from Table 2 which the compositional effect reflects. In Panel A, only age compositional changes are considered based on effects in Figures E2, while in Panels B-D age effects are considered based on the composition of separate groups documented in Figure E5.

by age across a range of characteristics, we also observe little evidence to suggest that there is a broad change in the composition of births across categories such as whether an individual comes from families in which the mother has high education, or in couples with relatively better matches as proxied by the age difference between partners or whether fathers are observed on the birth certificate. While we observe relatively little evidence of changes in birth timing in this sample, we can consider the relevance of this as a potential mechanism by determining what any compositional changes would imply in terms of birth outcomes if we expressed these changes in terms of mother fixed effect estimates documented in Figure E2. Specifically, we calculate the difference in mean birth outcomes expected if births among treated individuals actually had the same composition as untreated outcomes.[20] We observe that such compositional effects are small. If we

[20]For example, in Table E4, we observe that treated individuals are very slightly more likely to be 17 years old, slightly less likely to be 19 years old and so forth. From Figure E2, we know that 17 year olds on average have births which weigh around 3200 grams, while 19 year olds have births that weigh around 3250 grams. We can thus back out the compositional change by mapping the distribution of ages above the cut-off into the distribution of ages below the cut-off. This is:

$$\text{Compositional Effect} = \sum_{a=15}^{25} \left\{ [\Pr(Age = a|Treat = 1) - \Pr(Age = a|Treat = 0)] \times \text{Health at Age } a \right\},$$

consider only changes in birth timing, Table 4 suggests that compositional changes account for at most 1.2% of the documented RDD effect. However, if we consider changes in composition using mother fixed effect models separately by education and partnership status, we observe that compositional changes can account for between around 0.3 to 4.5% of the documented RDD effect (Table 4, panel D) with the lower range of these effects being on measures where treatment effects were observed to be substantial. Taken together, the results in this section thus suggest that changes in timing are likely of second order importance compared with changes in selection into giving birth.

We cannot directly assess the relevance of concerns related to external validity. Indeed, as we emphasise in the introduction, it is important to view our results as indicative of the effect of policy receipt on our sample of individuals aged up to the 25 years. Nevertheless, we can draw some relevant information from the findings documented in this section. We have shown that the effect of compositional changes within the time-window studied is relatively minor. We have also shown that stocks of health at birth begin to worsen as mother's age increases beyond 25 years. Finally, we have noted in the previous section that fertility increased by around 0.048 births as a result of treatment receipt. Taken together, these facts may suggest that births occurring to untreated women above the age of 25 are unlikely to revert the estimates documented here. However, this is simply suggestive evidence, and whether such a fertility channel remains valid beyond the age of 25 is an open question.

5.3 Marginal versus Average Effects

The model laid out in Section 4 and the analysis of channels of intergenerational transmission raises the question of what can be said about the *direct* intergenerational transmission channel, or the 'structural' component laid out in Section 4. We have argued above that our evidence shows that there are marginal births which occur among treated women but not among untreated women, and these marginal births obscure our ability to observe the direct effect of treatment receipt.

We can nevertheless ask what these marginal births may look like, and whether this lines up with the evidence discussed previously. To do so, we will consider 'marginal' births as births which would occur if the mother receives treatment, but not if the mother marginally misses out. If we denote the share of marginal births among treated mothers as a proportion μ, with the remaining portion $(1-\mu)$ of births occurring to always-mothers who would give birth whether they received treatment or not. If we denote as α^M and α^A the causal effect of treatment receipt on marginal and always-mothers conditional on having children, we

where Health at age a is estimated from mother FE models.

can decompose the estimated effect of treatment as:

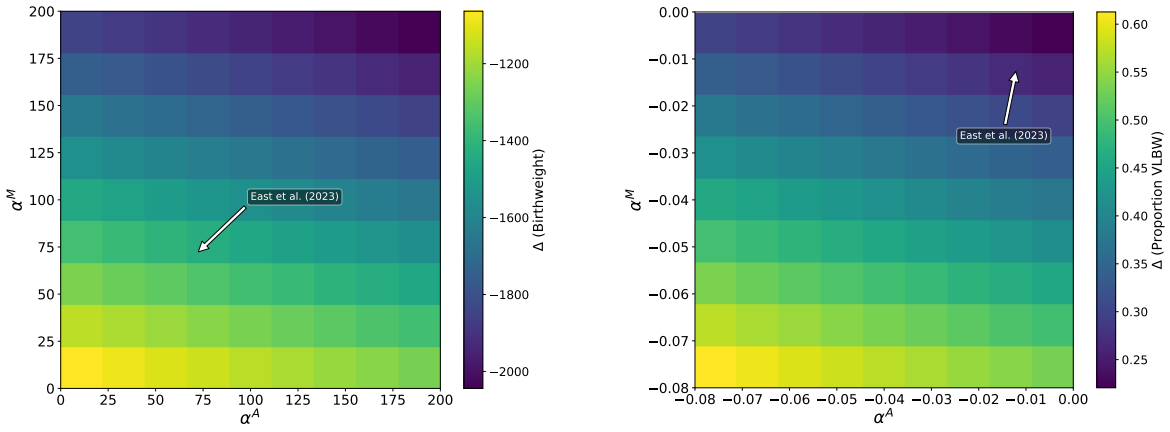
$$\alpha = \mu\alpha^M + (1-\mu)\alpha^A + \mu\Delta, \tag{7}$$

where Δ represents a selection term. Specifically, this selection effect refers to the difference in average outcomes between marginal and always-treated mothers when they are untreated. We provide a formal explanation of (7) in Appendix E.3, along with a graphical explanation in Figure E6. Indeed, this derivation is more generally true for any regression discontinuity estimate or treatment effect with an extensive and intensive margin.

To understand what we gain from this decomposition, we can consider what value of Δ would rationalise treatment effects if $(\alpha^M, \alpha^A) = 0$, which is presumably a lower bound for the direct effect of improved health at birth on next generation's health. For example, consider our estimated effect of treatment receipt on birth weight of -217 grams. Also, consider that from Table E2, we can estimate the proportion of marginal births as $\frac{0.048}{0.235} = 0.20$ (the 'extra' births not observed among untreated women). If we substitute these values into (7), this suggests that $\Delta = \frac{-217}{0.20} = -1064$ grams. This is illustrative as it suggests that the difference in birth weights between marginal and always treated individuals is substantial, which is what we would expect if treatment receipt was allowing second generation births with the most delicate health stocks to be live births, and hence appear in data. It is also interesting in that it is *not* consistent with an explanation of smaller changes in birth weight as a result of differences in birth timing (*i.e.* it suggests an extensive rather than intensive margin explanation). More generally however, we can ask what Δ would have to be to rationalise our treatment effects if (α^M, α^A) vary. We consider this below in Figure 8 in which the value of Δ is plotted as a colour gradient assuming a range of possible values of α^M and α^A.

This figure suggests that results line up with our previous posited mechanism of fertility selection in the second generation. Even if the true intergenerational impacts of early life treatment receipt are quite large, values of Δ are credible if they represent the births which would have not survived in the second generation. For example, if we assume that the effects of this policy are similar to those of Medicaid as estimated by East et al. (2023), this would imply that marginal births would have needed to be around 1,500 grams lighter than always births had treatment not been received. Considering that the mean second generation birth weight is 3,157 grams, this suggests that the births which are saved in the second generation are those which are around 2,000 grams. Similar intuition can be drawn in considering rates of VLBW. For the estimates from East et al. (2023), around 30% of marginal births would be expected to themselves be VLBW. Once again, this lines up with the posited mechanism in which the marginal births which are saved are precisely those

Figure 8: Values of Δ which rationalise α



(a) Birth weight **(b)** VLBW

Notes: Colours presented in heat maps represent values for Δ which would rationalise treatment effects of $\alpha = -217.7$ (panel A), and $\alpha = 0.045$ (panel B), given specific values of α^A and α^M plotted on the x and y axis respectively. The point indicated by the white arrow refers to the estimate of the intergenerational returns to prenatal Medicaid eligibility in which no fertility effect of the reform was observed.

which have the worst health stocks.

6 Conclusion

In this paper we document a long shadow to public policies, with the impacts of intensive medical care at birth found to have appreciable impacts as much as a quarter of a century later, and to be transmitted by recipients across generations. We document that unlike a large literature showing virtuous impacts of policies and positive shocks when passed from mothers to their children, we observe that children of treated mothers have worse measured birth outcomes such as low birth weight.

In examining policy mechanisms, we find that this unexpected result owes to selection into child birth as a result of early life medical care. Girls who are born weighing just below 1,500 grams, and who receive intensive early life investments are much *more* likely to go on to have their own birth. We find that policy receipt in generation 1 makes individuals more able to give birth, but on average give birth to babies with weaker stocks of health at birth. We document a new stylised fact clearly linking birth weight to future fertility (both within and outside of the bandwidth considered in this study), and also make clear that this relationship is modifiable, as treatment receipt is observed to increase rates of birth by around 20% around VLBW cut-offs. In many settings where concerns exist about low or below-replacement rate fertility, this suggests another social return to early life investments. Indeed both this finding and back-of-the-envelope

cost-benefit analysis suggests that while in this setting second generation health at birth is observed to be worse on average as a result of policy receipt (owing to the fertility channel), total policy benefits far exceed costs.

It is interesting to consider why these results are different to those generally found in the extant literature. One reason may owe to external validity. During the period under study Chile was a middle-income country, and much of the literature examining intergenerational spillovers is drawn from a high-income setting. What's more, data demands in following individuals across generations are high, and our analysis is limited only to mothers aged up to 25. Thus, our evidence must be viewed through this lens, as we are unable to observe transmission among older mothers, or from fathers to their children. However, there is reason to believe that these findings may be relevant to similar groups in broader settings. Estimated impacts of this specific early life treatment on both first generation health and later life educational outcomes has been shown to be comparable in Chile to Norway, Denmark, and the US (Bharadwaj et al., 2013; Daysal et al., 2022; Almond et al., 2010), suggesting some external validity in other outcomes in this domain. The relevance of an extensive (fertility) and intensive margin (health at birth) policy response seems to be key in explaining differences with existing results. The only other study examining intergenerational spillovers of health policies of which we are aware (East et al., 2023) is focused on a specific low-income group of recipients and finds no fertility response, and subsequent positive intergenerational spillovers. A similar lack of fertility response has been documented in Nilsson (2017) when considering alcohol consumption, suggesting that the degree to which extensive and intensive margin policy effects come into play and why is a relevant question to consider.

These findings have implications related to the ways downstream public health and welfare policies are defined following initial policy receipt. While a number of influential papers show that the impacts of early life treatments are unambiguously positive for their recipients (Bharadwaj et al., 2013; Almond et al., 2010; East et al., 2023), suggesting the need for compensatory policies for individuals who marginally miss out on such policies, our results also point to the importance of reinforcing investments, at least when considering the second generation of the original policy recipients in this sample.

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